

ANSWERS & EXPLANATIONS

GENERAL STUDIES (P) TEST – 3792 (2023)

Q 1.C

- Temperature is the most ecologically relevant environmental factor. The average temperature on land varies seasonally, decreases progressively from the equator towards the poles and from plains to the mountain tops. It ranges from subzero levels in polar areas and high altitudes to >50 degrees in tropical deserts in summer. There are, however, unique habitats such as thermal springs and deep-sea hydrothermal vents where average temperatures exceed 100 degrees.
- Temperature affects the kinetics of enzymes and through basal metabolism, activity and other physiological functions of the organism. **A few organisms can tolerate and thrive in a wide range of temperatures (they are called eurythermal), but, a vast majority of them are restricted to a narrow range of temperatures (such organisms are called stenothermal).** The levels of thermal tolerance of different species determine to a large extent their geographical distribution.
- Next to temperature, water is the most important factor influencing the life of organisms. In fact, life on earth originated in water and is unsustainable without water. Its availability is so limited in deserts that only special adaptations make it possible to live there.
- The productivity and distribution of plants is also heavily dependent on water. For aquatic organisms the quality (chemical composition, pH) of water becomes important.
- The salt concentration (measured as salinity in parts per thousand), is less than 5 in inland waters, 30-35 the sea and > 100 percent in some hypersaline lagoons.
- **Some organisms are tolerant of a wide range of salinities (euryhaline) but others are restricted to a narrow range (stenohaline). Many freshwater animals cannot live for long in sea water and vice versa because of the osmotic problems, they would face.**
- **Hence, both statements are not correct.**

Q 2.D

- A comprehensive central legislation was enacted in 1972 called the Wildlife (Protection) Act for providing special legal protection to our wildlife and to the endangered species of fauna in particular.
- **Wild Life (Protection) Act, 1972** provides a legal framework for the protection of various **species of wild animals and plants**, the management of their habitats, and the regulation and control of trade in wild animals, plants and products made from them. The Act also lists schedules of plants and animals that are afforded various degrees of protection and monitoring by the government. The Act provides for punishment for violation of its provisions. The Act also provides for **forfeiture of any equipment, vehicle or weapon** that is used for committing wildlife offence(s). A provision was incorporated in 2002 for the **constitution of the National Board for Wildlife**
- **Wildlife Protection Act, 1972** forbids hunting and poaching of wild animals. But under **section 11 of the Act, hunting of wild animals is permitted** in the following cases.
 - **The Chief Wild Life Warden may**, if he is satisfied that any wild animal specified in Schedule I of the Wildlife Protection Act, 1972 has become dangerous to human life or is so disabled or diseased as to be beyond recovery, **permit any person to hunt such animal. Hence, statement 1 is correct.**
 - The **killing or wounding in good faith of any wild animal in defence of oneself** or of any other person shall not be an offence. **Hence, statement 2 is correct.**
 - Any wild animal **killed or wounded in defence of any person shall be Government property. Hence, statement 3 is correct.**

Q 3.A

- **Allen's rule states that mammals from colder climates generally have shorter ears and limbs to minimise heat loss. A smaller body surface area helps animals in colder regions stay warm by slowing down the loss of body heat.** It is named after American biologist Joel Asaph Allen who proposed it in his 1877 paper "The influence of physical conditions in the genesis of species".
- **Bergmann's rule** states that organisms at higher latitudes should be larger and thicker than those closer to the equator to better conserve heat.
- **Hence, option (a) is the correct answer.**

Q 4.B

- Photosynthetically active radiation designates the spectral range of solar radiation from 400 to 700 nanometers that photosynthetic organisms are able to use in the process of photosynthesis. This spectral region corresponds more or less with the range of light visible to the human eye.
- **Statement 1 is not correct:** Except for the deep sea hydro-thermal ecosystem, sun is the only source of energy for all ecosystems on Earth. **Of the incident solar radiation less than 50 per cent of it is photosynthetically active radiation (PAR).**
- **Statement 2 is correct:** Plants capture only 2-10 per cent of the PAR and this small amount of energy sustains the entire living world.

Q 5.C

- **Aquatic Ecosystem:** Lakes and ponds are divided into three different "zones" which are usually determined by depth and distance from the shoreline.
- **Littoral Zone:** The **topmost zone near the shore of a lake or pond is the Littoral zone.** This zone is the warmest since it is shallow and can absorb more of the sun's heat. It sustains a fairly diverse community, which can include several species of algae (like diatoms), rooted and floating aquatic plants, grazing snails, clams, insects, crustaceans, fishes, and amphibians. The vegetation and animals living in the littoral zone are food for other creatures such as turtles, snakes, and ducks.
- **Limnetic Zone:** The near-surface open water surrounded by the littoral zone is the limnetic zone. The limnetic zone is **well-lighted (like the littoral zone) and is dominated by plankton, both phytoplankton and zooplankton.** plankton are small organisms that play a crucial role in the food chain. Without aquatic plankton, there would be few living organisms in the world, and certainly no humans. A variety of freshwater fish also occupy this zone. **Hence statement 1 is correct.**
- **Profundal zone:** The deep-water part of the lake or pond is called the profundal zone. This zone is **much colder and denser than the other two.** Little light penetrates all the way through the limnetic zone into the profundal zone. The **fauna are heterotrophs**, meaning that they eat dead organisms and use oxygen for cellular respiration. **Hence statement 2 is correct.**

Q 6.C

- Ecologists and evolutionary biologists have proposed various hypotheses to account for the greater biological diversity in the Tropics.
 - Speciation is generally a function of time, unlike temperate regions subjected to frequent glaciations in the past, tropical latitudes have remained relatively undisturbed for millions of years and thus, had a long evolutionary time for species diversification. **Hence statement 1 is not correct.**
 - Tropical environments, unlike temperate ones, are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation and lead to a greater species diversity. **Hence statement 2 is not correct.**
 - There is more solar energy available in the tropics, which contributes to higher productivity; this in turn might contribute indirectly to greater diversity. **Hence statement 3 is correct.**

Q 7.A

- Parasitism has evolved in so many taxonomic groups from plants to higher vertebrates. Many parasites have evolved to be host-specific (they can parasitise only a single species of host) in such a way that both host and the parasite tend to co-evolve; that is, if the host evolves special mechanisms for rejecting or resisting the parasite, the parasite has to evolve mechanisms to counteract and neutralise them, in order to be successful with the same host species.
- In accordance with their life styles, parasites evolved special adaptations such as the loss of unnecessary sense organs, presence of adhesive organs or suckers to cling on to the host, loss of digestive system and high reproductive capacity. The life cycles of parasites are often complex, involving one or two intermediate hosts or vectors to facilitate parasitisation of its primary host.

- The human liver fluke (a trematode parasite) depends on two intermediate hosts (a snail and a fish) to complete its life cycle. The malarial parasite needs a vector (mosquito) to spread to other hosts. Majority of the parasites harm the host; they may reduce the survival, growth and reproduction of the host and reduce its population density.
- **Parasites that feed on the external surface of the host organism are called ectoparasites.** The most familiar examples of this group are the lice on humans and ticks on dogs. Many marine fish are infested with ectoparasitic copepods. Cuscuta, a parasitic plant that is commonly found growing on hedge plants, has lost its chlorophyll and leaves in the course of evolution. It derives its nutrition from the host plant which it parasitises. **Hence statement 1 is not correct.**
- **In contrast, endoparasites are those that live inside the host body at different sites (liver, kidney, lungs, red blood cells, etc.).** The life cycles of endoparasites are more complex because of their extreme specialisation. Their morphological and anatomical features are greatly simplified while emphasising their reproductive potential. **Hence statement 2 is not correct.**
- **Brood parasitism** in birds is a fascinating example of parasitism that lets the host incubate them. During the course of evolution, the eggs of the parasitic bird have evolved to resemble the host's egg in size and colour to reduce the chances of the host bird detecting the foreign eggs and ejecting them from the nest. **In brood parasitism in birds, the parasitic species does not obtain food,** nor does it depend on the host species for other functions, but only to incubate the eggs and feed the parasitic bird's chicks. **Hence statement 3 is correct.**

Q 8.C

- **Recent Context: In a crackdown on narcotic substances, the Railway Protection Force (RPF) recovered various narcotic products, including designer drugs, valued at over Rs 7.40 crore last month and arrested 165 people with NDPS.**
- The RPF has been empowered to conduct search, seizure and arrest under the Narcotic Drugs and Psychotropic Substances (NDPS) Act since April 2019 and has been actively participating in the efforts of the government to restrict this illegal trade.
- **In order to bring attention to the menace of NDPS, a month-long pan-India drive against the smuggling of narcotics through rail was launched under the code name Operation "NARCOS" in the month of June-2022.**
- Under this operation RPF has performed excellently against the carriers/transporters of Narcotics product through Indian Railways. RPF intensified its checks in trains and in identified black spots across the country in coordination with Narcotics Control Bureau (NCB) and other Law Enforcement Agencies (LEAs) to target drug peddlers involved in this illegal trade.
- **Hence option (c) is the correct answer.**

Q 9.D

- **DECOMPOSITION:**
 - When a plant, animal or insect dies it breaks into tiny pieces and eventually, these small pieces become part of the soil. This process is known as decomposition. It is the process where organic substances break down into a simpler matter. bodies of any living organism start decomposing shortly after their death. Animals such as worms help in decomposing the organic matter.
 - **Decomposers** break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition. **The important steps in the process of decomposition are fragmentation, leaching, catabolism, humification and mineralisation. Hence, option (d) is the correct answer.**
 - **Detritivores** (e.g., earthworm) break down detritus into smaller particles. This process is called **fragmentation**.
 - By the process of leaching, water soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called as **catabolism**. It is important to note that all the above steps in decomposition operate simultaneously on the detritus.
 - **Humification and mineralisation** occur during decomposition in the soil. Humification leads to accumulation of a dark coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate. Being colloidal in nature it serves as a reservoir of nutrients. The humus is further degraded by some microbes and release of inorganic nutrients occur by the process known as **mineralisation**.

Q 10.C

- **Bishnois, the tribal community of Rajasthan are considered the first environmentalist in the world.** Of the 29 tenets of their religion, eight tenets are dedicated to the preservation of biodiversity and animal husbandry without killing. Felling of trees has been prohibited. **Bishnois remove living insects from the firewood before burning.**
 - **They never cut trees and wait for trees to die on their own or fall down during storms.** They are non-violent with self-control and their dead bodies are buried in the woods despite being Hindu, so as to ensure that mineralisation of the organic matter provides nutrition to the growing trees. **Wildlife like Chinkara, Blackbucks and Great Indian Bustard move freely in their settlements.**
- **Gonds:** The Gonds, the largest Adivasi Community in India are of Dravidian origin and can be traced to the pre-Aryan era. The Gonds or the Koiture are a heterogeneous group spreading over large areas from the Godavari gorges in the south to the Vindhya Mountains in the north. In Madhya Pradesh, they inhabited the dense forests of the Vindhyas, Satpura and Mandla in the Narmada region of the Amarkantak range for centuries.
- **Bhils:** They are considered as one of the oldest tribe in India. Once they were the ruler in parts of Rajasthan, Gujarat, Malwa, Madhya Pradesh and Bihar. Bhils could be identified as one of the Dravidian racial tribe of Western India and belong to Austroloid group of tribes. They speak a language of Dravidian origin.
- **Baigas:** They are an ethnic group found in central India primarily in the state of Madhya Pradesh, and in smaller numbers in the surrounding states of Uttar Pradesh, Chhattisgarh and Jharkhand. The name Baiga means "sorcerer- medicine man".
- **Hence, option (c) is the correct answer.**

Q 11.D

- The **first National Wildlife Action Plan (NWAP) was adopted in 1983**, based upon the decision taken in the meeting of the Indian Board for Wildlife held in 1982. The plan had outlined the strategies and action points for wildlife conservation. **Hence, statement 1 is not correct.**
- Now, **India's National Wildlife Action Plan (NWAP) for the period 2017-2031** is released. The five components under the plan are:
 - strengthening and promoting the integrated management of wildlife and their habitats
 - adaptation to climate change and promoting integrated sustainable management of aquatic biodiversity in India; promoting eco-tourism
 - nature education and participatory management
 - strengthening wildlife research and monitoring of development of human resources in wildlife conservation
 - enabling policies and resources for conservation of wildlife in India.
- Due to the rapid decline in wildlife population, the Government of India during 1952 had constituted an advisory body designated as the **Indian Board for Wildlife (IBWL).**
 - The Wildlife (Protection) Act, 1972 was subsequently amended during 1991 and last during 2002. As per the amendment of the Act in 2002, a provision was incorporated for the constitution of the National Board for Wildlife, replacing the Indian Board for Wildlife. The **National Board for Wildlife is chaired by the Prime Minister.** **Hence, statement 2 is not correct.**

Q 12.C

- **Sacred Groves** are the tracts of virgin forests that are left untouched by the local inhabitants and are protected by the local people due to their culture and religious beliefs. Sacred groves are relic vegetation of once-dominant flora. They are repositories of our rich biodiversity; they are also the last bastion where the rich culture and the customs of the indigenous people are still preserved.
- A sacred grove usually consists of a dense cover of vegetation including climbers, herbs, shrubs and trees, with the presence of a village deity and is mostly situated near a perennial water source.
- They are called with different names in different states:
 - Sarna in Bihar
 - Dev Van in Himachal Pradesh
 - **Devarakadu in Karnataka. Hence pair 3 is correctly matched.**
 - **Kavu in Kerala. Hence pair 2 is not correctly matched.**
 - Dev in Madhya Pradesh
 - Devarahati or Devarai in Maharashtra
 - Lai Umang in Manipur
 - Law Kyntang or Asong Khosi in Meghalaya

- Oran in Rajasthan
- Kovil Kadu or Sarpa Kavu in Tamil Nadu
- **Than in Assam. Hence pair 1 is correctly matched.**

Q 13.C

- **Ex-situ conservation means conservation of wildlife outside its natural habitat.** The conservation takes place in captivity under man's supervision. Some times the populations of species may decline or may become extinct due to genetic or environmental factors such as inbreeding, habitat loss, disease and over-exploitation. In such cases in-situ conservation may not prove to be effective and a species can be protected from becoming extinct only through maintaining them in artificial conditions under human care. Such measures are included under Ex-Situ Conservation Methods.
- Examples are: **Botanical gardens, zoo, aquariums, parks, agricultural research centre, forest research centres, etc.**
 - A botanical garden is a garden dedicated to the collection, cultivation, preservation and display of an especially wide range of plants, which are typically labelled with their botanical names.
 - A zoo is a facility in which animals are housed within enclosures, cared for, displayed to the public, and in some cases bred for conservation purposes.
- A **Biosphere Reserve** is a unique and representative ecosystem of terrestrial and coastal areas which are internationally recognised within the framework of UNESCO's Man and Biosphere (MAB) Programme. The Biosphere Reserve aims at achieving the three objectives - conservation, development and research. Biosphere Reserve is an example of **In-Situ Conservation method**.
- **Hence, option (c) is the correct answer.**

Q 14.A

- **Conformer organism: Conformers engage in behavior to regulate their temperature, such as basking in the sun for warmth or retreating underground or into water to cool. Some aquatic animals even alter their salinity to match that of the environment around them. An overwhelming majority of animals and nearly all plants cannot maintain a constant internal environment. Their body temperature changes with the ambient temperature.** In aquatic animals, the osmotic concentration of the body fluids changes with that of the ambient air, and water osmotic concentration.
- Many simply 'sweat it out and resign themselves to suboptimal performance in hot summer months. Thermoregulation is energetically expensive for many organisms. This is particularly true for small animals like shrews and hummingbirds. At night, hummingbirds lower their body temperature and metabolism drastically by dropping into an energy-saving state of inactivity called torpor.
- Heat loss or heat gain is a function of surface area. Since small animals have a larger surface area relative to their volume, they tend to lose body heat very fast when it is cold outside; then they have to expend much energy to generate body heat through metabolism. This is the main reason why very small animals are rarely found in polar regions.
- During the course of evolution, the costs and benefits of maintaining a constant internal environment are taken into consideration. Some species have evolved the ability to regulate, but only over a limited range of environmental conditions, beyond which they simply conform.
- Regulators are often referred to as warm-blooded animals. Examples of this class – Mammals and birds (peacock).
- **Hence option (a) is the correct answer.**

Q 15.C

- **Green chemistry:** It is also called as sustainable chemistry, is an area of chemistry and chemical engineering focused on the design of products and processes that minimize or eliminate the use and generation of hazardous substances.
- **Green Chemistry in day-to-day Life:**
 - **Dry Cleaning of Clothes: Tetrachloroethene** ($\text{Cl}_2\text{C}=\text{CCl}_2$) was earlier used as solvent for dry cleaning. The compound contaminates the ground water and is also a suspected carcinogen. The process using this compound is now being replaced by a process, where liquefied carbon dioxide, with a suitable detergent is used. Replacement of halogenated solvent by liquid CO_2 will result in less harm to ground water. These days **hydrogen peroxide (H_2O_2)** is used for the purpose of bleaching clothes in the process of laundry, which gives better results and makes use of lesser amount of water.

- **Bleaching of Paper:** Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H₂O₂) with suitable catalyst, which promotes the bleaching action of hydrogen peroxide, is used. Hence statement 2 is correct.
- **‘Green Solution’ to Clean Turbid Water:** Powder of kernel of tamarind seeds has been found to be an effective material to make municipal and industrial waste water clean. It is non-toxic, biodegradable and cost-effective material. This powder is usually discarded as agricultural waste. The present practice is to use alum to treat such water. **It has been found that alum increases toxic ions in treated water and can cause diseases. Hence statement 1 is correct.**

Q 16.A

- For any species, the minimal requirement is one more species on which it can feed. Even a plant species, which makes its own food, cannot survive alone; it needs soil microbes to break down the organic matter in the soil and return the inorganic nutrients for absorption.
- Even in minimal communities, many interactive linkages exist, although all may not be readily apparent. Interspecific interactions arise from the interaction of populations of two different species. They could be beneficial, detrimental or neutral (neither harm nor benefit).

Species A	Species B	Name of Interaction
+	+	Mutualism
-	-	Competition
+	-	Predation
+	-	Parasitism
+	0	Commensalism
-	0	Amensalism

- **Hence, option (a) is the correct answer.**
- Interspecific **Competition**, in ecology, is a form of competition in which individuals of different species compete for the same resources in an ecosystem. It adversely affects both organisms.
- **Mutualism** is an association between organisms of two different species in which each benefit. Mutualistic arrangements are most likely to develop between organisms with widely different living requirements. One example of a mutualistic relationship is that of the bee and the flower. In this mutualistic relationship, the bees get to eat, and the flowering plants get to reproduce.
- **Predation** is a biological interaction where one organism, the predator, kills and eats another organism, its prey. Predators help in maintaining species diversity in a community by reducing the intensity of competition among competing prey species. Examples are leopards, tigers etc.
- **Parasitism** is a close relationship between species, where one organism, the parasite, lives on or inside another organism, the host, causing it some harm, and is adapted structurally to this way of life. Roundworm, malaria parasite, many bacteria, fungi and viruses are common parasites of humans.
- **Commensalism** is a type of relationship between two living organisms in which one organism benefits from the other without harming it. The relationship between trees and epiphytic plants is an example of commensalism.
- **Amensalism** is an association between organisms of two different species in which one is inhibited or destroyed and the other is unaffected. Example - A large tree shades a small plant and retards the growth of the small plant. The small plant has no effect on the larger tree.

Q 17.B

- **Recently, a ‘Statue of Peace’ of Swami Ramanujacharya made of marble was inaugurated in Srinagar by the Union Home Minister. Situated in the Sonwar region of Srinagar, this incredible statue symbolizes peace in the valley and in the country.**
- **Ramanujacharya visited Kashmir in the 11th Century to get an important manuscript called Bodhayana Vritti, a treatise on the Brahma Sutras. The Bodhayana Vritti had the reputation of being the most authoritative explanation of the Brahma Sutras.**
- **Ramanujacharya is revered as a Vedic philosopher and social reformer. He traveled across India, advocating equality and social justice. He brought the treasure of Vedic literature to the doorsteps of the common man, became the flag bearer of the Vaishnav sect, and advocated the philosophy of Vishishtadvaita, qualified monism.**

- While dispelled the Mayavada concept, that the world is illusionary **he Became the preceptor of the Bhakti movement** and the source for all other Bhakti Schools of thought.
- Ramanuja's most **important writings include his commentary on the Vedanta Sutras (the Sri Bhasya, or "True Commentary"), and his commentary on the Bhagavad-Gita (the Gitabhasya, or "Commentary on the Gita").**
- He was an inspiration for mystic poets like Kabir, Meerabai, Annamacharya, Bhakta Ramdas, Thyagaraja, and many others. He is credited with the initiation of the concept that Nature and her resources like Water, Air, Soil, Trees, etc., are sacred and should be protected from pollution.
- **Hence option (b) is the correct answer.**

Q 18.B

- **Types of Pollutants: On the basis of origin in the environment, pollutants are classified into the following three types:**
 - **Natural Pollutants:** The pollutants which come out from natural sources such as forest fires started by lightning, dispersal of pollen, soil erosion, volcanic eruptions, volatile organic compounds from leaves and trees, decomposition of organic matter and natural radioactivity, etc. are natural pollutants.
 - **Primary Pollutants:** A primary pollutant is a harmful substance that directly enters the air as a result of human activities. For example, when coal, oil, natural gas or wood is burnt, carbon dioxide and carbon monoxide are formed, and automobiles contribute a large share of carbon monoxide. All these gases enter the atmosphere. Another important pollutant is sulphur dioxide (SO₂) nitrogen, hydrocarbons and suspended particulate matter.
 - **Secondary Pollutants:** Secondary pollutants result from the harmful chemical reaction between two or more air components. For example, sulphur dioxide, the primary pollutant reacts with oxygen in the atmosphere to form the secondary pollutant, sulphur trioxide (SO₃) ($2\text{SO}_2 + \text{O}_2 = 2\text{SO}_3$). The sulphur trioxide can then react with water vapour in the air to form droplets of sulphuric acid (H₂SO₄), another secondary pollutant. Troposphere Ozone (O₃) is another secondary pollutant.
 - **On the basis of existence in nature, pollutants are also divided into two categories:**
 - **Quantitative pollutants:** These substances exist in nature but when their concentration or quantity increases up to a critical level then they act as pollutants: e.g. Carbon dioxide, Nitrogen oxide, etc.
 - **Qualitative pollutants:** These pollutants do not exist in the environment. They are introduced into the environment through man's activities. e.g. **pesticides (DDT), chemical fertilizers, herbicides** etc.
- Hence, option (b) is the correct answer.**

Q 19.C

- Radioactive wastes are the most hazardous of all. For example, some of the components can retain half of their dangerous levels even one million years later after production. The storing and reprocessing are further complicated by the long half-life of the radioactive materials in the nuclear waste.
- Radioactive materials such as uranium and radium possess highly unstable atomic nuclei whose disintegration results in radiation emission which may be highly injurious. **During nuclear tests, radioactive dust may encircle the globe at altitudes of 3000 meters or more.** This dust often comes down to earth as rain. Some of it percolates down through the soil into ground, water reservoirs or is carries into rivers and streams. **Hence statement 1 is correct.**
- **Radioactive wastes are generated from X-ray machines in hospitals and airports, nuclear energy industry which include substances used in cooling and storing nuclear fuel from reactors in power stations and submarines. Hence statement 2 is correct.**

Q 20.A

- A constant input of solar energy is the basic requirement for any ecosystem to function and sustain itself. Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. The rate of biomass production is called productivity.
- Primary productivity depends on the plant species inhabiting a particular area. It also depends on a variety of environmental factors, availability of nutrients and the photosynthetic capacity of plants. Therefore, it varies in different types of ecosystems. It can be divided into gross primary productivity (GPP) and net primary productivity (NPP). **The gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis. Hence, statement 1 is correct.**
- **A considerable amount of GPP is utilized by plants in respiration. Gross primary productivity minus respiration losses (R), is the net primary productivity (NPP).** Net primary productivity is the available biomass for the consumption of heterotrophs (herbivores and decomposers). Secondary

productivity is defined as the rate of formation of new organic matter by consumers. **Hence, statement 2 is correct.**

- **The annual net primary productivity of the whole biosphere is approximately 170 billion tonnes (dry weight) of organic matter. Of this, despite occupying about 70 percent of the surface, the productivity of the oceans is only 55 billion tonnes. Hence, statement 3 is not correct.**

Q 21.C

- **People for the Ethical Treatment of Animals (PETA)** was founded in 1980 and is dedicated to establishing and defending the rights of all animals. PETA operates under the simple principle that animals are not ours to experiment on, eat, wear, use for entertainment, or abuse in any other way. PETA informs policymakers and the public about animal abuse and promotes the kind treatment of animals.
- **PETA is an international non-profit charitable organization based in Norfolk, Virginia, with entities worldwide. Hence, option (c) is the correct answer.**
- PETA opposes speciesism, a human-supremacist worldview, and focuses its attention on the four areas in which the largest numbers of animals suffer the most intensely for the longest periods of time: in laboratories, in the food industry, in the clothing trade, and in the entertainment business.

Q 22.A

- Water Hyacinths are the beautiful mauve-coloured flowers found on very appealingly-shaped floating plants in water bodies. These plants which were introduced into India for their lovely flowers have caused havoc by their excessive growth by causing blocks in our waterways. They grow faster than our ability to remove them. These are plants of water hyacinth (*Eichhornia crassipes*), the world's most problematic aquatic weed, also called 'Terror of Bengal'. They grow abundantly in eutrophic water bodies, and lead to an imbalance in the ecosystem dynamics of the water body.
- Domestic sewage primarily contains biodegradable organic matter, which readily decomposes – thanks to bacteria and other micro-organisms, which can multiply using these organic substances as substrates and hence utilise some of the components of sewage. It is possible to estimate the amount of biodegradable organic matter in sewage water by measuring **Biochemical Oxygen Demand (BOD)**.
- **Water Chestnuts:** Despite their name, water chestnuts are not nuts but instead are aquatic vegetables grown in marshes and ponds that are popular ingredients for Asian dishes. Water chestnuts, also known as Chinese water chestnuts, are tuberous, aquatic vegetables that grow underwater in marshes, ponds, paddy fields, and shallow lakes.
- **Water Lilies:** Water lily is a freshwater flowering plants, native to the temperate and tropical parts of the world. Water lilies provide food for fish and wildlife but sometimes cause drainage problems because of their rapid growth.
- **Pothos** is a low maintenance perennial evergreen houseplant in the Araceae (arum) family. Desired for its glossy, green or variegated leaves on cascading stems. It only grows 6 to 8 feet as a horizontal groundcover, but the trailing, climbing vines can grow as long as 40 feet. This feature makes it well-suited for hanging baskets.
- **Hence, option (a) is the correct answer.**

Q 23.C

- **Desert Ecosystem:** Deserts make up about one-fifth of the surface of the earth and occur where rainfall is less than, 50 cm/year. The major deserts are the Sahara, the Arabian, Kalahari, and Namib of Africa, the Atacama of South America, the Mojave, Sonoran, and Chihuahuan of North America, and the Australian deserts. Besides, there are the Thar Desert in western India and Pakistan and the Gobi Desert of Mongolia.
- **Features of Desert Ecosystems include:**
 - **The soil in the desert is coarse. They are coarse because there is less chemical weathering. The finer dust and sand particles are blown elsewhere, leaving heavier pieces behind. Canopy in most deserts is very rare. Plants are mainly ground-hugging shrubs and short woody trees. Hence, statement 1 is correct.**
 - In the deserts, days are very hot and nights are cold. So the plants and animals are **exposed to both types of extremes.**
 - The predominant plants found in the deserts are succulent species with waxy surfaces, such as cacti, which can conserve water for longer periods of time, or deciduous shrubs, also with thick waxy leaves. **Hence, statement 3 is correct.**
 - The animal types in the deserts are few. These include burrowing and nocturnal rodents, reptiles, insects, and arachnids (Scorpions, spiders, etc.). These animals adapt themselves to the temperature extremes and desiccation of desert air by living beneath the surface during the day and moving out

only at night. Most of them have remarkable water conservation adaptations. **Hence, statement 2 is not correct.**

- **Most of the animals living in the deserts are nocturnal** (active only at night). Common animals are camels, foxes, kangaroos, rats, squirrels, etc.
- **Many of the insects have waxy coats and reduced water loss through the cuticles.**
- The amount and kinds of plants vary according to where a desert is located. **Short grasses can be found in nearly all deserts. Desert plants include sagebrush, creosote bushes, and cacti.**

Q 24.B

- **Project Tiger: It was a project launched in India in 1973 for the protection and breeding of tigers so as to increase their population.** Different tiger reserves were established in different parts of the country under this project. It was by 1997 that the number of tiger reserves, rose up to 27 and the number of tigers rose up to 1500, in India. **Project Tiger is administered by the National Tiger Conservation Authority.** National Tiger Conservation Authority (NTCA) is a statutory body of the Ministry, with an overarching supervisory/coordination role, performing functions as provided in the Wildlife (Protection) Act, 1972.
- **Project Elephant: It was a project launched in India as a centrally sponsored scheme to protect the population of wild elephants. This project was launched in 1992 simultaneously in 12 states.** Inspite of this project, the population of wild elephants is coming down due to massive deforestation and disruption of their migration routes by human activities.
 - It Indian elephant *Elephas maximus* occurs in the central and southern Western Ghats, North-east India, eastern India and northern India and in some parts of southern peninsular India. It is included in **Schedule I of the Indian Wildlife (Protection) Act, 1972** and in **Appendix I of the Convention on International Trade in Endangered Species of Flora and Fauna (CITES).**
- **Project Crocodile: It is a project formulated to protect crocodiles from the impending danger of their extinction in 1975.** Under the project crocodile breeding centres were established in different places of the country in order to increase their population. Today, more than 30 centres are in operation across the country.
- **Hence, option (b) is the correct answer.**

Q 25.D

- Plant-based derivatives are widely available and are potential sources for many bioactive components such as vitamins, minerals and natural antioxidants, etc.
- **Some of the substances that we extract from plants on a commercial scale are nicotine, caffeine, quinine, strychnine, opium, etc**
- **Nicotine is a plant alkaloid, found in the tobacco plant. It is a highly addictive chemical compound.**
- **Caffeine is a natural chemical with stimulant effects. It occurs naturally in more than 60 plants.** Caffeine works by stimulating the central nervous system, heart, muscles, and the centers that control blood pressure.
- **Quinine is a drug obtained from cinchona bark that is used chiefly in the treatment of malaria, an infection caused by the protozoan parasite Plasmodium, which is transmitted to humans by the bite of various species of mosquitoes.**
- **Strychnine is an alkaloid that is found in Strychnos nux-vomica (Loganiaceae). It is a medium-sized, deciduous tree, with fairly straight and cylindrical bole and dark-grey or yellowish-grey bark with minute tubercles.**
- **Hence, option (d) is the correct answer.**

Q 26.C

- **AIR POLLUTION AND ITS CONTROL**
 - There are several ways of removing particulate matter; the most widely used of which is the **electrostatic precipitator**, which can remove over 99 per cent particulate matter present in the exhaust from a thermal power plant. It has electrode wires that are maintained at several thousand volts, which produce a corona that releases electrons. These electrons attach to dust particles giving them a net negative charge. The collecting plates are grounded and attract the charged dust particles. The velocity of air between the plates must be low enough to allow the dust to fall. **Hence, statement (1) is correct.**
 - **A scrubber can remove gases like sulphur dioxide.** In a scrubber, the exhaust is passed through a spray of water or lime. Particulate matter are very very small and are not removed by precipitators. According to Central Pollution Control Board (CPCB), particulate size 2.5 micrometers or less in

diameter (PM 2.5) are responsible for causing the greatest harm to human health. These fine particulates can be inhaled deep into the lungs and can cause breathing and respiratory symptoms, irritation, inflammations and damage to the lungs and premature deaths. **Hence, statement (2) is not correct.**

- Automobiles are a major cause for atmospheric pollution at least in the metro cities. **Catalytic converters, having expensive metals namely platinum-palladium and rhodium as the catalysts, are fitted into automobiles for reducing emission of poisonous gases.** As the exhaust passes through the catalytic converter, unburnt hydrocarbons are converted into carbon dioxide and water, and carbon monoxide and nitric oxide are changed to carbon dioxide and nitrogen gas, respectively. Motor vehicles equipped with catalytic converter should use unleaded petrol because lead in the petrol inactivates the catalyst. **Hence, statement (3) is correct.**

Q 27.D

- **Major abiotic factors influencing the life of organisms include:**
 - **Temperature:** Temperature is the most important ecologically relevant environmental factor. **A few organisms can tolerate and thrive in a wide range of temperatures (they are called eurythermal), but, a vast majority of them are restricted to a narrow range of temperatures (such organisms are called stenothermal).** The levels of thermal tolerance of different species determine to a large extent their geographical distribution.
 - **A poikilotherm is an animal whose internal temperature varies considerably. Poikilotherms have to survive and adapt to environmental stress. They are organisms (such as a frog) with a variable body temperature that tends to fluctuate with and is similar to or slightly higher than the temperature of its environment. Hence option (d) is the correct answer.**
 - **Water:** Water is another the most important factor influencing the life of organisms. The productivity and distribution of plants are also heavily dependent on water. For aquatic organisms the quality (chemical composition, pH) of water becomes important. The salt concentration (measured as salinity in parts per thousand), is less than 5 in inland waters, 30-35 in the sea, and > 100 in some hypersaline lagoons. Some organisms **are tolerant of a wide range of salinities (euryhaline)** but **others are restricted to a narrow range (stenohaline).** Many freshwater animals cannot live for long in seawater and vice versa because of the osmotic problems, they would face.
 - **NOTE:** Ecosystems are capable of maintaining their state of equilibrium. They can regulate their own species structure and functional processes. This capacity of the ecosystem of self-regulation is known as **homeostasis**. When an organism tries to maintain a constant internal (within the body) environment that permits all biochemical reactions and physiological functions to proceed with maximal efficiency and thus, enhance the overall 'fitness' of the species. This process is known as Homeostasis.
 - **Hence option (d) is the correct answer.**

Q 28.C

- Recently, all 198 members of the United Nations **adopted a landmark declaration at the United Nations' Ocean Conference in Lisbon to scale up science-based and innovative actions and address the ocean emergency of habitat loss, ocean acidification and ecosystem degradation. This declaration has been declared the Lisbon Declaration.**
- The Lisbon declaration 'Our ocean, our future, our responsibility,' called on governments to **do more to prevent, reduce, and eliminate marine plastic litter - including single-use plastics and microplastics** - by contributing to comprehensive life-cycle approaches, encouraging recycling and environmentally sound waste management.
- The **declaration welcomed the decision made at the fifth UN Environment Assembly held in Nairobi, Kenya, earlier this year to establish an intergovernmental negotiating committee to develop a legally binding instrument on plastic pollution.** The Member States gavelled a historic resolution to forge the agreement by 2024.
- The declaration acknowledged the need to establish greater, better-managed Marine Protected Areas. It noted the voluntary commitments by more than 100 Member States to conserve or protect at least 30 per cent of the global ocean within Marine Protected Areas and other area-based conservation measures by 2030.
- **Hence option (c) is the correct answer.**

Q 29.B

- **Eutrophication**

- Lakes receive their water from surface runoff and along with its various chemical substances and minerals.
- Over periods spanning millennia, aging occurs as the lakes accumulate mineral and organic matter and gradually, get filled up.
- The nutrient-enrichment of the lakes promotes the growth of algae, aquatic plants, and various fauna. This process is known as natural eutrophication.
- The process of eutrophication takes place due to the introduction of nutrients and chemicals through the discharge of domestic sewage, industrial effluents, and fertilizers from agricultural fields.
- Similar nutrient enrichment of lakes at an accelerated rate is caused by human activities and the consequent aging phenomenon is known as cultural eutrophication.
- On the basis of their nutrient content, lakes are categorized as Oligotrophic (very low nutrients), Mesotrophic (moderate nutrients), and Eutrophic (highly nutrient-rich).
- A vast majority of lakes in India are either eutrophic or mesotrophic because of the nutrients derived from their surroundings or organic wastes entering them.

- **Eutrophication and Algal Bloom**

- **Eutrophic water body: It is a body of water rich in nutrients and so supports a dense plant population, the decomposition of which kills animal life by depriving it of oxygen. Hence option (b) is the correct answer.**
- Eutrophication is the response to the addition of nutrients such as nitrates and phosphates naturally or artificially, fertilizing the aquatic ecosystem.
- Phytoplankton (algae and blue-green bacteria) thrive on the excess nutrients and their population explosion covers almost the entire surface layer. This condition is known as an algal bloom.

- **Mechanism**

- Phytoplankton is photosynthetic during the daytime adding oxygen to the aquatic ecosystem.
- But during the night, they consume far more oxygen as they respire aggressively.
- i.e. Algal blooms accentuate the rate of oxygen depletion as the population of phytoplankton is very high.
- The primary consumers like small fish are killed due to oxygen deprivation caused by algal blooms.
- The death of primary consumers adversely affects the food chain.
- Further, more oxygen is taken up by microorganisms during the decomposition process of dead algae, plants, and fishes.
- The new anaerobic conditions (absence of oxygen) were created to promote the growth of bacteria such as *Clostridium botulinum* which produces toxins deadly to aquatic organisms, birds, and mammals.
- Algal blooms can be any color, but the most common ones are red or brown. These blooms are commonly referred to as red or brown tides.

- **Effects of Eutrophication**

- Loss of freshwater lakes: Eutrophication eventually creates a detritus layer in lakes and produces a successively shallower depth of surface water.
- Eventually, the water body is reduced into a marsh whose plant community is transformed from an aquatic environment to a recognizable terrestrial environment.
- Algal Blooms restrict the penetration of sunlight resulting in the death of aquatic plants and hence restricting the replenishment of oxygen.

- **New species invasion:**

- Eutrophication may cause the ecosystem competitive by transforming the normal limiting nutrient to abundant levels. This cause shifting in species composition of the ecosystem.
- Loss of coral reefs: Occurs due to a decrease in water transparency (increased turbidity).
- Affects navigation due to increased turbidity; creates color (yellow, green, red), smell, and water treatment problems; increases biomass of inedible toxic phytoplankton, benthic and epiphytic algae, and bloom of gelatinous zooplankton.
- Harmful Algal Blooms
- Most algal blooms are not harmful, but some produce toxins. These are known as Harmful Algal Blooms (HABs).
- Toxicity: Some algal blooms when died or eaten, release neuro & hepatotoxins which can kill aquatic organisms & pose a threat to humans. E.g. Shellfish poisoning.
- HAB events adversely affect commercial and recreational fishing, tourism, and valued habitats, creating a significant impact on local economies and the livelihood of coastal residents.

Q 30.A

- **Effects of Pollution on Animals:** The effects of pollution on animals are more or less similar to the effects of pollution on human beings. However, being top consumers in most of the food chains, human beings are not directly exposed to pollutants, though they receive accumulated pollutants in greater concentrations and hence are prone to greater risks as compared to secondary consumers. The effects of pollution on animals are summarized below.
 - The pollution of air causes different types of diseases in the respiratory systems of animals. Numerous species of insects and birds are killed due to polluted air.
 - High pitched sounds produced due to explosions, fires of bullets, etc. kill a large number of birds and insects. Terrorist camps in forests frighten wild animals during fire testing and explosion practices.
 - The **high-pitched sound is also produced by Supersonic planes**. The effect produced in the atmosphere by the flight of a supersonic plane and heard by our ears is called a “**sonic boom**”. The term sonic boom is commonly used to refer to the shocks caused by the supersonic flight of an aircraft. Sonic booms generate enormous amounts of sound energy, sounding much like an explosion.
 - Millions of fish are killed every year by a wide variety of different pollutants from municipal and industrial sources.
 - Pesticides applied in agriculture are lethal to numerous types of animals. The **application of the Carbaryl pesticide kills honey bees** which are most important for the pollination of crop plants. This pesticide is lethal to fish and birds also. **Toxaphene, another pesticide, causes deformities in the backbones of fishes.** The **Dichloro Diphenyl Trichloroethane (DDT)** and its breakdown products **tend to retard the reproductive capacities of birds and cause thinning of their egg- cells. Hence option (a) is the correct answer.**
 - In the marine environment, some ornithologists estimate that up to 250,000 birds are killed each year by the effect of oil spilled on the sea- surface.
 - **Water and mud contaminated with pesticides kill thousands of swallows due to Pharyngitis** caused when their throats are exposed to pesticides when they go for collecting mud for building up their nests.
 - Hundreds of **cows die every year due to ingestion of food containing polythene bags** thrown along roadsides and on open grounds.
 - Large numbers of **fish are killed in lakes and ponds due to Eutrophication** as their waters get contaminated by plant- nutrients and sewage containing organic substances. The biological decomposition of these substances causes a scarcity of oxygen which kills fish and amphibians.

Q 31.C

- **Indian Forest Act, 1927** is an act to consolidate the law relating to forests, the transit of forest produce and the duty leviable on timber and other forest produce.
- According to the act, The State Government is empowered to notify forest land as the property of the Government and appoint officers to take care of forest land. **Hence, statement 1 is correct.**
- The act gives many powers to the State Government which include
 - State Government can declare the trees in a protected forest as reserved. **Hence, statement 2 is not correct.**
 - The State Government may constitute any forest-land or waste-land which is the property of Government, or over which the Government has proprietary rights, or to the whole or any part of the forest-produce of which the Government is entitled, a reserved forest.
 - State Government may prohibit the quarrying of stone, burying of lime or charcoal or removal of any forest produce in a forest.
 - State Government may assign to a village community the rights of government over any land that has been constituted as a reserve forest. **Hence, statement 3 is correct.**
- It states that the practice of shifting cultivation shall in all cases be deemed a privilege subject to control, restriction and abolition by the State Government.

Q 32.B

- **Productivity:**
 - **Primary production** is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis. It is expressed in terms of weight (gm^{-2}) or energy (kcal m^{-2}).
 - The rate of biomass production is called Productivity. It is expressed in terms of $\text{gm}^{-2} \text{ yr}^{-1}$ or (kcal m^{-2}) yr^{-1} to compare the productivity of different ecosystems. It can be divided into Gross Primary Productivity (GPP) and Net Primary Productivity (NPP).

- Gross primary productivity of an ecosystem is the rate of production of organic matter during photosynthesis. A considerable amount of GPP is utilised by plants in respiration.
- **Gross primary productivity** minus respiration losses (R), is the net primary productivity (NPP). $GPP - R = NPP$ Net primary productivity is the available biomass for the consumption to heterotrophs (herbivores and decomposers).
- **Secondary productivity** is defined as the rate of formation of new organic matter by consumers.
- **Primary productivity** depends on the plant species inhabiting a particular area. It also depends on a variety of environmental factors, availability of nutrients and photosynthetic capacity of plants. Therefore, it varies in different types of ecosystems. **Hence, option (b) is the correct answer.**
- The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter. Of this, despite occupying about 70 per cent of the surface, the productivity of the oceans are only 55 billion tons.

Q 33.A

- **Recently, scientists at the US Lawrence-Berkeley Lab have developed a fuel from a bacterium that packs more energy than even the rocket fuels in use today.** Data from simulation show that the fuel has energy density values exceeding 50 megajoules a litre, compared with 32 MJ for petrol and 35 MJ for RP-1, a kerosene-based rocket fuel. **The scientists have named the new fuel 'POP-FAME', for polycyclopropanated fatty acid methyl ester.**
- **POP-FAME's molecular structure closely resembles Syntin's.** At the heart of the fuel's structure is the 'three-carbon' ring — a triangle with a carbon atom at each vertex. (Each carbon atom combines with two other carbon atoms and two other elements, mostly hydrogen.) This structure is called cyclopropane; they hold potential energy in their bonds.
- **Syntin Fuel:** During the 1960s, a petroleum-based rocket fuel Syntin was developed by the Soviet Union. This fuel was used to launch several Soyuz rockets, successfully in the 1970s. however, its manufacturing was stopped due to high costs and the unpleasant process involved. It was produced through a series of synthetic reactions with toxic by-products and explosive & unstable intermediate.
- POP-FAME is said to have higher energy densities than Syntin, which means even a small quantity of the fuel can pack considerable energy, making it ideal rocket fuel.
- **Hence option (a) is the correct answer.**

Q 34.B

- Decomposers break down complex organic matter into inorganic substances like carbon dioxide, water and nutrients and the process is called decomposition. Dead plant remains such as leaves, bark, flowers and dead remains of animals, including fecal matter, constitute detritus, which is the raw material for decomposition.
- The important steps in the process of decomposition are fragmentation, leaching, catabolism, humification and mineralisation. Detritivores (e.g., earthworm) break down detritus into smaller particles. This process is called fragmentation.
- By the process of leaching, water-soluble inorganic nutrients go down into the soil horizon and get precipitated as unavailable salts.
- **Bacterial and fungal enzymes degrade detritus into simpler inorganic substances. This process is called catabolism.** It is important to note that all the above steps in decomposition operate simultaneously on the detritus. **Hence, statement 1 is correct.**
- **Humification and mineralization occur during decomposition in the soil. Humification leads to the accumulation of a dark-coloured amorphous substance called humus that is highly resistant to microbial action and undergoes decomposition at an extremely slow rate. Hence, statement 3 is correct.**
- Being colloidal in nature it serves as a reservoir of nutrients. The humus is further degraded by some microbes and the release of inorganic nutrients occurs through the process known as mineralization.
- **Decomposition is largely an oxygen-requiring process. The rate of decomposition is controlled by the chemical composition of detritus and climatic factors. In a particular climatic condition, the decomposition rate is slower if detritus is rich in lignin and chitin, and quicker if detritus is rich in nitrogen and water-soluble substances like sugars. Hence, statement 2 is not correct.**
- Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes. Warm and moist environments favour decomposition whereas low temperature and anaerobiosis inhibit decomposition resulting in the build-up of organic materials.

Q 35.D

- Waste is generated when materials become useless to the owner and they wish to discard them. It is a mid-step between generation and collection of waste. It may not be needed if the people are dumping waste directly into pits. In cases, where there are heaps of waste, the storage facilities must be provided like big containers, bins, drums, shallow pits or walled areas. Enacted in 1986, the Environment Protection Act aims to establish a sufficient protection system for the environment. It gives the power to the central government to regulate all forms of waste and to tackle specific problems that may present themselves in different regions of India. **However, it does not define 'waste'. Hence statement 1 is not correct.**
- **Wastes generated during mining operations, excavations etc. are called mining wastes.** They include tailings, slag heaps, debris etc. Some mining processes use large volumes of chemicals or liquids, which add to the waste generated. **In mining, tailings are the materials left over after the process of separating the valuable fraction from the uneconomic fraction (gangue) of an ore. Hence statement 2 is not correct.**
- The Environment Ministry revised the Solid Waste Management Rules in 2016. Some of salient features of the rules include:
 - The Rules are now applicable beyond Municipal areas and extend to urban agglomerations, census towns, notified industrial townships, areas under the control of Indian Railways, airports, airbase, Port and harbour, defence establishments, special economic zones, State and Central government organizations, places of pilgrims, religious & historical importance.
 - The source segregation of waste has been mandated to channelize the waste to wealth by recovery, reuse and recycle.
 - Responsibilities of Generators have been introduced to segregate waste in to three streams, Wet (Biodegradable), Dry (Plastic, Paper, metal, wood, etc.) and domestic hazardous wastes (diapers, napkins, empty containers of cleaning agents, mosquito repellents, etc.) and handover segregated wastes to authorized rag-pickers or waste collectors or local bodies.
 - Integration of waste pickers/ ragpickers and waste dealers/ Kabadiwalas in the formal system should be done by State Governments, and Self Help Group, or any other group to be formed.
 - No person should throw, burn, or bury the solid waste generated by him, on streets, open public spaces outside his premises, or in the drain, or water bodies. Generator will have to pay 'User Fee' to waste collector and for 'Spot Fine' for Littering and Non-segregation.

Q 36.A

- **Recently, the Prime Minister unveiled the national emblem that has been placed atop the new Parliament building in a special ceremony.**
- The national emblem is an adaption of the Lion Capital of Ashoka at Sarnath. **The statue is a three-dimensional emblem showing four lions, mounted back to back, on a circular abacus.**
- Its base has an **elephant, a galloping horse, a bull, and a lion separated by intervening wheels** over a bell-shaped lotus and a **Dharma Chakra or the Wheel of Law carved on it.**
- The emblem also has a phrase inscribed below the abacus **in the Devanagari script that says: "Satyameva Jayate"** It's a quote taken from Mundaka Upanishad which says Truth Alone Triumphs.
- **Hence option (a) is the correct answer.**

Q 37.A

- **Smog:** The word smog is derived from smoke and fog. This is the most common example of air pollution that occurs in many cities worldwide. There are two types of smog:
 - **Classical smog** occurs in cool humid climates. It is a mixture of smoke, fog and sulphur dioxide. Chemically it is a reducing mixture and so it is also called reducing smog.
 - **Photochemical smog occurs in warm, dry and sunny climate. Hence, statement 2 is not correct.**
 - The main components of the photochemical smog **result from the action of sunlight on unsaturated hydrocarbons and nitrogen oxides produced by automobiles and factories. Ground-level ozone constitutes one of the major components of photochemical smog.** Photochemical smog has a high concentration of oxidizing agents and is, therefore, called oxidizing smog.
 - **Hence, statement 1 is correct.**
- **Effects of photochemical smog:**
 - **The common components of photochemical smog are ozone, nitric oxide, acrolein, formaldehyde and peroxyacetyl nitrate (PAN).** Photochemical smog causes serious health problems. Both ozone and PAN act as powerful eye irritants.

- Ozone and nitric oxide irritate the nose and throat and their high concentration causes a headache, chest pain, dryness of the throat, cough, and difficulty in breathing. Photochemical smog leads to the cracking of rubber and extensive damage to plant life. It also causes corrosion of metals, stones, building materials, rubber and painted surfaces.
- **How can photochemical smog be controlled?**
 - Many techniques are used to control or reduce the formation of photochemical smog. If we control the primary precursors of photochemical smog, such as NO₂ and hydrocarbons, and the secondary precursors such as ozone and PAN, the photochemical smog will automatically be reduced.
 - Usually, catalytic converters are used in automobiles, which prevent the release of nitrogen oxide and hydrocarbons into the atmosphere. Certain plants e.g., Pinus, Juniperus, Quercus, Pyrus and Vitis can metabolise nitrogen oxide, and therefore, their plantation could help in this matter.

Q 38.D

- Corals are massive wave-resistant structures built largely by coral and consisting of skeletal and chemically precipitated material. These are made by polyps i.e. animals belonging to the class **Coelenterata**.
- In the Indian seas, coral reefs are found in the **Gulf of Mannar, Pak-bay, Lakshadweep (atolls), Gulf of Kutch, and Andaman and Nicobar sea areas**. The health of coral reefs has been declining over the past several decades. The pollution of seawater, destructive fishing practices, algal blooms, and eutrophication, coral diseases and coral bleaching are some serious causes of depletion of coral reefs.
- The whitening of coral colonies due to the loss of symbiotic zooxanthellae from the tissues of polyps is called as **Coral Bleaching**. **Zooxanthellae** are unicellular algae that provide colour to corals. They also provide food to corals and their deaths lead to the breakdown of the symbiotic relationship between them. This breakdown of symbiotic relationship causes starvation and resultant deaths of coral polyps. This condition exposes the white calcium carbonate skeletons of the coral colony.
- **There are a number of stresses or environmental changes that may cause bleaching. These causes include disease, excess shade, increased levels of ultraviolet radiation, sedimentation, pollution, salinity changes, and increased temperatures.**
- Other causes of coral bleaching include
 - **Increased exposure to ultraviolet (UV) radiation**
 - **Large amounts of stormwater from heavy rains flooding the reef**
 - **The exposure of coral to certain chemicals or diseases**
 - **Sediments such as sand or dirt covering the coral**
 - **Excess nutrients such as ammonia and nitrate from fertilizers and household products entering the reef ecosystem.** (The nutrients might increase the number of zooxanthellae in the coral, but it is possible that the nutrient overload increases the susceptibility of coral to diseases.)
- **Hence, option (d) is the correct answer.**

Q 39.B

- **National Forest Policy (NFP) 1988** envisages that **33%** of the geographical area should be under forest or tree cover. **Hence, statement 1 is not correct.**
- Salient features of the National Forest Policy are :
 - Maintenance of environmental stability and restoration of ecological balance.
 - Conservation of natural heritage.
 - Checking soil erosion and denudation.
 - Increasing substantially the forest/tree cover.
 - Meeting the requirements of fuelwood, fodder, NTFP and small timber.
 - Increasing productivity of forests.
 - Encouraging efficient utilisation of forest produce and maximising substitution.
 - Creating a massive people's movement especially with the involvement of women.
- It provides for **compensatory afforestation** and Wastelands and degraded lands have been suggested to be utilized for afforestation and other plantation purposes. **Hence, statements 2 and 3 are correct.**
 - It states that the diversion of forest land for any non-forest purpose should be subject to the most careful examinations by specialists from the standpoint of social and environmental costs and benefits. Construction of dams and reservoirs, mining and industrial development and expansion of agriculture should be consistent with the need for the conservation of trees and forests. Projects which involve such diversion should at least provide in their investment budget, funds for **regeneration/compensatory afforestation**.

Q 40.D

- A **National park** is an area which is strictly reserved for the betterment of the wildlife and where **human activities like forestry, grazing or cultivation are not permitted**. A National Park is defined by state government via notification. The state government can fix and alter boundaries of the National Parks with prior consultation and approval with National Board of Wildlife. **Hence statement 1 is not correct.**
- **Jim Corbett National Park: Established in 1936, this is India's first national park.** It is named after the legendary naturalist and conservationist Jim Corbett. Located at the Himalayas' foothills (**Uttarakhand**), near the popular hill-station of **Nainital**, it is famous for being home to a large number of tigers, the highest among any Indian national park. The park is spread over the picturesque landscapes of Pauri Garhwal, Almora, and Nainital. **Hence statement 2 is not correct.**
- A wildlife sanctuary is a protected area that is reserved for the conservation only of wildlife- animals and plant species. Human activities like harvesting or timber collection of minor forest products and private ownership rights are allowed. **Wildlife Sanctuary is species oriented** such as Pitcher plant, Great Indian Bustard.
- Biosphere Reserves are divided into three zones namely, core zone or natural zone, buffer zone and transition zone.
 - **Core zone:** It is an undistributed and legally protected ecosystem. ie, in this **no human activity is permitted.**
 - **Buffer zone:** The area that surrounds the core zone is called buffer zone. In this zone human activity is permitted.
 - **Transition zone:** The outermost region of the biosphere is termed transition zone. **It is the area of active cooperation between reserve management and the local people. Hence statement 3 is not correct.**

Q 41.B

- **The Air (Prevention and Control of Pollution) Act was enacted in 1981** and amended in 1987 to provide for the prevention, control and abatement of air pollution in India.
- **The Water (Prevention and Control of Pollution) Act was enacted in 1974** to provide for the prevention and control of water pollution, and for the maintaining or restoring of the wholesomeness of water in the country. The Act was amended in 1988. The Water (Prevention and Control of Pollution) Cess Act was enacted in 1977, to provide for the levy and collection of a cess on water consumed by persons operating and carrying on certain types of industrial activities. This cess is collected with a view to augment the resources of the Central Board and the State Boards for the prevention and control of water pollution constituted under the Water (Prevention and Control of Pollution) Act, 1974. The Act was last amended in 2003.
- **The Environment (Protection) Act was enacted in 1986** with the objective of providing for the protection and improvement of the environment. It empowers the Central Government to establish authorities [under section 3(3)] charged with the mandate of preventing environmental pollution in all its forms and to tackle specific environmental problems that are peculiar to different parts of the country. The Act was last amended in 1991
- **The Forest (Conservation) Act, 1980**, came into force to address deforestation. Though the Indian Forest Act has been in force since 1927, it was geared to allow the colonial British administration to control the extraction of timber and not aimed at preserving forests or addressing deforestation. There have been at least two major amendments to the FCA — in 1988 and 1996.
- **Hence, option (b) is correct.**

Q 42.C

- **Antibiosis** is an association between two or more organisms that is detrimental to at least one of them. In this the production of secretions by an organism that is harmful to others. **In this type of relationship, none of the species benefits. Hence statement 1 is correct.**
- For example, some species of **blue-green algae that grows** in ponds produce toxic substances that kill fishes as well as cattle that drink water. In marine waters, the population of some microbes popularly known as red tide causes the destruction of fish and other animals. **Hence statement 2 is correct.**

Q 43.B

- **KINGDOM FUNGI**
 - The fungi constitute a unique kingdom of **heterotrophic organisms**. They show a great diversity in morphology and habitat. One must have seen fungi on moist bread and rotten fruits. White spots seen on mustard leaves are due to a parasitic fungus.

- **Some unicellular fungi, e.g., yeast** are used to make bread and beer. Other fungi cause diseases in plants and animals; wheat rust-causing *Puccinia* is an important example. **Some are the source of antibiotics, e.g., *Penicillium*. Hence, option (b) is the correct answer.**
- **Fungi are cosmopolitan and occur in air, water, soil, and on animals and plants.** They prefer to grow in warm and humid places. With the exception of yeasts which are unicellular, fungi are filamentous. Their bodies consist of long, slender thread-like structures called hyphae.
- **The network of hyphae is known as mycelium.** Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called coenocytic hyphae. Others have septae or cross walls in their hyphae.
- **The cell walls of fungi are composed of chitin and polysaccharides.** Most fungi are heterotrophic and absorb soluble organic matter from dead substrates and hence are called saprophytes. Those that depend on living plants and animals are called parasites.
- **They can also live as symbionts** – in association with algae as lichens and with roots of higher plants as mycorrhiza. Reproduction in fungi can take place by vegetative means – fragmentation, fission, and budding.
- **Asexual reproduction is by spores called conidia** or sporangiospores or zoospores, and sexual reproduction is by oospores, ascospores, and basidiospores. The various spores are produced in distinct structures called fruiting bodies.
- The sexual cycle involves the following three steps:
 - The fusion of protoplasts between two motile or non-motile gametes is called plasmogamy.
 - The fusion of two nuclei is called karyogamy.
 - Meiosis in zygotes results in haploid spores.
- When a fungus reproduces sexually, **two haploid hyphae of compatible mating types come together and fuse.** In some fungi, the fusion of two haploid cells immediately results in diploid cells (2n).
- However, **in other fungi (ascomycetes and basidiomycetes), an intervening dikaryotic stage ($n + n$, i.e., two nuclei per cell) occurs;** such a condition is called a dikaryon and the phase is called dikaryophase of fungus. Later, the parental nuclei fuse, and the cells become diploid. The fungi form fruiting bodies in which reduction division occurs, leading to the formation of haploid spores.

Q 44.D

- **Flora:** Based on the type of forests, the flora also varies from forest to forest.
 - **In Tropical Forests** one square kilometer may contain as many as 100 different tree species. Trees are 25-35 m tall, with shallow roots, mostly evergreen, with large dark green leaves. Plants such as **orchids, bromeliads, vines (lianas), ferns, mosses, and palms are present in tropical forests.**
 - Flora is characterized by 3-4 tree species per square kilometer in the **Temperate forests.** Trees are distinguished by broad leaves which shed their leaves annually during the dry season. They include such species as **oak, hickory, beech, hemlock, maple, basswood, cottonwood, elm, willow, and spring-flowering herbs. Hence option (d) is the correct answer.**
 - **Taiga forests** consist mostly of cold-tolerant evergreen conifers with **needle-like leaves, such as pine, fir, and spruce.**

Q 45.C

- **What is Biological Oxygen Demand (BOD)?**
 - Biological Oxygen Demand is defined as the amount of dissolved oxygen required by aerobic microorganisms to break down the organic materials in a sample of water at a specific temperature & timeframe. In simpler words, The amount of oxygen needed by biological organisms such as bacteria in a given water sample for a breakdown of organic matter by oxidation process is called the Biochemical Oxygen Demand.
- **Why is this method used?**
 - BOD is the biological method used for the measurement of the total amount of dissolved oxygen (DO) used by microbes in the biological process of metabolizing organic molecules present in water.
 - The total amount of oxygen gas present in the water is called dissolved oxygen (DO). The non-compound oxygen present in water may either be a by-product of the photosynthesis of the aquatic plants or the dissolved atmospheric oxygen gas.
- **Significance**
 - BOD measures the amount of oxygen utilized by microorganisms for the process of decomposition of the organic matter in the water bodies.

- **It symbolizes the amount of organic pollution present in an aquatic ecosystem. Hence, statement 1 is correct.**
- Also regulates the chemical oxidation (COD) of inorganic matter.
- Used in sewage treatment or wastewater treatment to destroy and decay organic wastes through the aerobic organisms.
- Determines the amount of organic matter present in soils, sewages, sediment, garbage, sludge, etc.
- Detects the rate of respiration in living beings.
- Used in the medicinal & pharmaceutical industries to test the oxygen consumption of cell cultures.
- Biological oxygen demand can be used as an indicator of the level of environmental pollution. For instance, the higher the levels of organic matter (in polluted water bodies or sewage), the greater the BOD.
- Consequently, **a high Biological Oxygen Demand level means the amount of dissolved oxygen available for other marine organisms such as fish is low. This causes the mortality of fish and other aquatic creatures. Hence, statement 2 is correct.**
- From an ecological perspective, wastewater treatment plants aim to lower their BOD before discharging the effluents into a water body.

Q 46.D

- **Recently, the World Bank released 'The Global Findex Report 2021'.** The Global Findex surveys over 125,000 adults in 123 economies during Covid-19 to better understand how people use formal and informal financial services and digital payment. The report highlights that Large shares of the global population without formal banking (130 million and 230 million, respectively) live in India and China because of their size. **Hence pair 3 is correctly matched.**
- **Recently, The UN-Habitat World Cities Report 2022** highlighted that rapid urbanization in India was delayed temporarily due to the covid-19 pandemic. This new report calls for greater commitment by national, regional, and local governments, and encourages the further adoption of innovative technologies and urban living concepts such as the “15-minute city.” Popularized in Paris, Melbourne, and other places, it envisages that residents can meet most of their daily needs within a 15-minute journey. **Hence pair 1 is correctly matched.**
- Hyderabad has made it to the list of top 20 sustainable cities as per the **Asia Pacific (APAC) Sustainability Index 2021**, along with Bengaluru, Delhi, and Mumbai. **Knight Frank**, in its latest report, Active Capital Asia-Pacific - Raising Capital in Uncertain Times, rated 36 cities based on urbanization pressure, climate risk, carbon emissions, and government initiatives and named the index the **Asia Pacific (APAC) Sustainability Index 2021. Hence pair 2 is not correctly matched.**
- The **European Intelligence Unit (EUI) recently released the Global Liveability Index 2022.** It ranked 173 cities on the basis of their liveability or living conditions, which was determined by five factors — stability, healthcare, culture and environment, education, and infrastructure. **The list, for the first time, included five Indian cities; Delhi, Mumbai, Chennai, Ahmedabad, and Bangalore.** When ranking, out of the five factors, the EUI attaches the highest weightage to stability and culture, and environment, which account for 25% each. It is followed by healthcare and infrastructure with 20% each and education with 10%.

Q 47.A

- **Joint Forest Management (JFM)** is an approach and program wherein state forest departments support **local forest dwelling and forest fringe communities** to protect and manage forests and share the costs and benefits from the forests with them. **Hence, statement 1 is correct.**
- Communities organize themselves into a JFM Committee to protect and manage nearby forests, guided by locally prepared byelaws and micro plans.
- The key element in JFM is that communities have the power to manage the use of forests by members and also exclude non-members.
- The benefits to them is direct access and control on the use and sale of most Non Timber Forest Produce and a share in the income from timber as well as other intangible benefits from local ecosystem services – like water recharge, pollination, wildlife habitat etc. Thus involvement of communities in conservation of forests and wildlife is of paramount interest.
- **Recognition in the National Forest Policy of 1988** that communities are central to forest protection and management led to the JFM Circular of June 1st 1990, and the subsequent 2000 and 2002 Guidelines, which provided the framework for state level rules, resolutions, and guidelines for JFM. The Panchayati Raj Act, PESA, and the FRA of 2006 further expanded the rights and responsibilities of local communities' vis-à-vis forests. **Hence, statement 2 is not correct.**

Q 48.C

- **Traditional water harvesting system:** In traditional methods of rainwater harvesting, water is stored and also used to recharge the groundwater. Some of the traditional methods of water conservation in different regions of India are:

Traditional Water Harvesting Systems	
States	Name of Traditional Water Harvesting System
Rajasthan	Kunds
Gujarat	Tankas
Maharashtra	Phad System
Andhra Pradesh	Tanks and Anicuts
Orissa	Katas
Karnataka	Teluparagis and Kere
Western Himalayas	Khatri or Kuhl
Gangetic plains	Baolis or Dighis
Central India	Johads
Western Ghats	Surangam
Eastern Ghats	Korambu
Ladakh	Zing

- Hence option (c) is the correct answer.

Q 49.D

- **Predation:**
 - Predation is the **interaction between prey and a predator**, where the energy flows from one organism to the other. The predator is the organism that feeds on other organisms called prey.
 - Besides acting as 'conduits' for energy transfer across trophic levels, predators play other important roles. They keep prey populations under control. But for predators, prey species could achieve very high population densities and cause ecosystem instability. When certain exotic species are introduced into a geographical area, they become invasive and start spreading fast because the invaded land does not have its natural predators.
 - Prey species have evolved various defenses to lessen the impact of predation. Some species of **insects and frogs are cryptically-colored (camouflaged) to avoid being detected easily by the predator**. Some are poisonous and therefore avoided by predators. The **Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body**. Interestingly, the butterfly acquires this chemical during its caterpillar stage by feeding on a poisonous weed. For plants, herbivores are predators.
 - Nearly 25 percent of all insects are known to be **phytophagous (feeding on plant sap and other parts of plants)**. The problem is particularly severe for plants because, unlike animals, they cannot run away from their predators. Plants, therefore, have evolved an astonishing variety of morphological and chemical defenses against herbivores.
 - **Thorns (Acacia, Cactus) are the most common morphological means of defense**. Many plants produce and store chemicals that make the herbivore sick when they are eaten, inhibit feeding or digestion, disrupt its reproduction or even kill it. For instance, the **weed Calotropis** growing in abandoned fields. The plant **produces highly poisonous cardiac glycosides** and that is why we never see any cattle or goats browsing on this plant.
 - To keep small predators at bay, many **plants have a mat of fine hairs on the surface of their leaves**. **Hence statement 1 is correct.**
 - Plant synthesize chemicals that are toxic to certain life forms. **Caffeine** is such a chemical; it has potent antibiotic and anti-fungal powers and causes sterility in several insects. Also, caffeine permeates the soil which surrounds the plants through the accumulation of fallen leaves and berries, thus inhibiting the growth of competing plants. **Hence statement 2 is correct.**
 - **A wide variety of chemical substances that we extract from plants on a commercial scale (nicotine, caffeine, quinine, strychnine, opium, etc.,) are produced by them actually as defenses against grazers and browsers.**
 - Plant structural traits such as **leaf surface wax, thorns or trichomes, and cell wall thickness/ and lignification** form the first physical barrier to feeding by the herbivores, and the secondary metabolites act as toxins and also affect growth, development, and digestibility reducers from the next barriers that defend the plant from subsequent attack. **Hence statement 3 is correct.**

Q 50.B

- The accelerated rates of species extinctions that the world is facing now are largely due to human activities. There are four major causes. 'The Evil Quartet' is the sobriquet used to describe them.
 - **Habitat loss and fragmentation:** This is the most important cause driving animals and plants to extinction. The most dramatic examples of habitat loss come from tropical rainforests. Besides total loss, the degradation of many habitats by pollution also threatens the survival of many species. When large habitats are broken up into small fragments due to various human activities, mammals and birds requiring large territories and certain animals with migratory habits are badly affected, leading to population declines.
 - **Ove-exploitation:** Many species extinctions in the last 500 years were due to overexploitation by humans. Presently many marine fish populations around the world are overharvested, endangering the continued existence of some commercially important species.
 - **Alien species invasions:** When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause the decline or extinction of indigenous species. The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake.
 - **Co-extinctions:** When a species becomes extinct, the plant and animal species associated with it in an obligatory way also become extinct. When a host fish species becomes extinct, its unique assemblage of parasites also meets the same fate. Another example is the case of a coevolved plant-pollinator mutualism where extinction of one invariably leads to the extinction of the other.
- Hence, option (b) is the correct answer.

Q 51.D

- **Sound Pollution:**
 - **In India, the Air (Prevention and Control of Pollution) Act came into force in 1981 but was amended in 1987 to include noise as an air pollutant.** Noise is an undesired high level of sound. We have got used to associating loud sounds with pleasure and entertainment not realizing that noise causes psychological and physiological disorders in humans.
 - Brief exposure to extremely high sound levels, 150 dB or more generated by the takeoff of a jet plane or rocket, may damage ear drums thus permanently impairing hearing ability.
 - Even chronic exposure to a relatively lower noise level in cities may permanently damage the hearing abilities of humans. Noise also causes sleeplessness, increased heartbeat, and altered breathing patterns, thus considerably stressing humans.
 - The reduction of noise in our industries can be affected by the use of sound-absorbent materials or by muffling noise.
- **What is Noise pollution according to Indian laws?**
 - Noise is anything that is an unwanted sound, according to the Central Pollution Control Board. Any unwanted sound that causes annoyance, irritation and pain to the human ear is termed 'noise'.
- **What is the acceptable level of noise?**
 - Rules have defined the acceptable levels of noise across areas during daytime and nighttime. Daytime comprises 6 am to 10 pm, and night is 10 pm to 6 am.
 - In commercial areas, noise limitation has been fixed at 65 dB and 55 dB during daytime and nighttime respectively. In residential areas, they are 55 dB and 45 dB during daytime and night respectively.
 - In industrial areas, the ceiling is fixed at 75dB and 70dB during daytime and nighttime, whereas in silence zones, the same is at 50dB and 40dB.
- **The Noise Pollution (Regulation and Control) Rules, 2000:**
 - **Section 2 (a) of the Air (Prevention and Control of Pollution) Act, 1981 considers noise as an 'air pollutant'.** According to reports, "air pollutant" is any solid, liquid, or gaseous substance, including noise, present in the atmosphere in such concentrations as to be or tend to be harmful to humans, other living creatures, plants, property, or the environment.
 - **Noise pollution and its sources are regulated under The Noise Pollution (Regulation and Control) Rules, 2000 under The Environment (Protection) Act, 1986. Hence, option (d) is the correct answer.**
 - Under the Act, there is a defined ambient acceptable noise level, restrictions on the use of loudspeakers, sound-emitting construction equipment, horns, bursting of crackers, and so on.

Q 52.D

- **GASEOUS POLLUTANTS:** The gaseous pollutants causing air pollution are numerous. Some of the important gaseous pollutants are mentioned below.
 - **Sulphur dioxide:** Sulphur dioxide (SO₂) and sulphur tri- oxide (SO₃) are produced largely by the combustion of coal and petroleum. These are also produced through smelting of ores of sulphide, copper, zinc, and lead; and decomposition of bio-mass. Some industries that emit sulphur dioxide are industries manufacturing sulphuric acid (H₂SO₄), oil refineries, fertilizer industries and paper industries.
 - **Hydrogen sulphide:** It is a colourless toxic gas which is produced from decaying vegetation and animal materials, particularly in shallow freshwater and marine environment. It also comes out from sulphur springs, volcanoes, coal pits and sewers.
 - **Carbon monoxide:** It is a poisonous gas which originates from the incomplete combustion of carbonaceous materials.
 - **Hydrogen fluoride:** It naturally comes out from volcanoes. However, it is produced from blast furnaces and industries concerned with the production of brick, tiles and superphosphates. It also comes out during the combustion of coal.
 - **Hydrogen chloride:** It comes out during the combustion of coal, paper, plastics and chlorinated hydrocarbons.
 - **Hydrocarbons:** Chemical compounds made of hydrogen and carbon, are called hydrocarbons. Methane, ethylene and aniline are three examples of hydrocarbons. Hydrocarbons are also produced due to incomplete combustion of fuels, automobile exhausts, petroleum refineries, burning of crop residues, cracking of natural gas in petrochemical plants etc. Major sources of release of hydrocarbons are organic matter, seepage from natural gas and oil fields and emissions of Volatile Organic Chemicals (VOCs).
 - **Methane:** It is the primary component of natural gas, and it warms the planet more than 80 times as quickly as a comparable volume of atmospheric CO₂ over a comparable amount of time. A major source of methane emissions is:

S.No.	Source	Methane release rate Per cent
1.	Natural wetlands	21.3
2.	Rice paddies	20.4
3.	Biomass burning	10.2
4.	Enteric fermentation	14.8
5.	Termites	7.4
6.	Landfills	7.4
7.	Gas drilling	8.3
8.	Coal mining	6.5
9.	Oceans	1.9
10.	Other sources	1.8
	Total	100.0

- Hence option (d) is the correct answer.
- **Ammonia:** This gas is principally generated through a refrigerator pre-cooler system of cold storage, manufacture of anhydrous ammonium fertilizers, nitric acid and domestic incineration etc. **The emission of ammonia causes bleaching of plant leaves, reduction of root and shoot growth, browning and softening of fruits, reduction in the rate of germination etc.**
- **Nitrogen Oxides:** Nitrogen oxide and Nitric Oxide are principal gaseous pollutants emitted through human activities. Nitrogen oxides have been reported to cause fading of colours of clothes, deterioration of nylon and cotton and corrosion of metals.
- **Tobacco Smoke:** Smoking cigarettes and allied things generate tobacco smoke. It is a potent pollutant in the closed atmosphere like buses; trains etc. It causes lung cancer and pulmonary and coronary heart diseases. Even passive smokers may be caught by pulmonary or coronary disease.

Q 53.B

- **The Atomic Energy Act of 1962** governs the use of radioactive materials and radiation-generating equipment. Under section 30 of this Act; the Central Government has issued **the Atomic Energy (Safe disposal of radioactive waste) Rules, 1987.**

- The **Atomic Energy Regulatory Board (AERB)** safety code includes salient recommendations on the disposal of radioactive waste. The Chairman of the AERB is designated as the Competent Authority to enforce the Atomic Energy (Safe disposal of radioactive waste) Rules, 1987. The employer, which may be the Head of the institution, shall obtain authorization from the Competent Authority for the disposal of radioactive waste either locally or through an authorized waste disposal agency. The Chairman, AERB may issue Surveillance procedures, codes, standards, and guides which elaborate the provisions of Rules for implementation.
- **Hence, option (b) is correct answer.**

Q 54.D

- **Recent context:** The Prime Minister recently launched new features of the ‘Prime Minister’s Employment Generation Programme (PMEGP). These include an increase in the maximum project cost to Rs 50 lakh (from Rs 25 lakh) for the manufacturing sector and Rs 20 lakh (from Rs 10 lakh) in the service sector.
- The recent changes also advocate for the **inclusion of applicants from Aspirational Districts and Transgender in the list of special category applicants for availing higher subsidies**. Also, handholding support is being provided to applicants/entrepreneurs through the engagement of banking, technical & marketing experts. **Hence statement 2 is correct.**
- It is a **central sector scheme** being administered by the Ministry of Micro, Small and Medium Enterprises, **being nodally implemented by Khadi and Village Industries Commission**. **Hence statement 3 is correct.**
- PMEGP seeks to facilitate the **generation of employment opportunities for youth across the country by assisting them in setting up micro-enterprises in non-farm sectors**. **Hence statement 1 is correct.**

Q 55.B

- **Recent context:** The Prime Minister has inaugurated the Digital India Week 2022 under the Digital India Programme, with the aim to strengthen Ease of Doing Business and Ease of Living.
- During the programme, PM launched multiple digital initiatives aimed at enhancing the accessibility of technology, streamlining service delivery to ensure ease of living and giving a boost to startups. The initiatives are as follows:
 - Digital India Bhashini: **Digital India BHASHINI is India’s Artificial Intelligence (AI) led language translation platform**. A Bhashini Platform will make Artificial Intelligence (AI) and Natural Language Processing (NLP) resources available to MSME (Medium, Small and Micro Enterprises), Startups and Individual Innovators in the public domain.
 - **Digital India GENESIS’ (Gen-next Support for Innovative Startups) is a national deep-tech startup platform to discover, support, grow and make successful startups in Tier-II and Tier-III cities of India**. **Hence pair 1 is correctly matched.**
 - **Meri Pehchaan: It is a National Single Sign On (NSSO) for One Citizen Login**. It is a user authentication service in which a single set of credentials provide access to multiple online applications or services. **Hence pair 3 is not correctly matched.**
 - My Scheme: It is a service discovery platform facilitating access to Government Schemes, which aims to offer a one-stop search and discovery portal where users can find schemes that they are eligible for.
 - **Indiastack: It is a global repository of key projects implemented under India Stack like Aadhaar, UPI (Unified Payment Interface), Digilocker, Cowin Vaccination Platform, Government e Marketplace, DIKSHA Platform and Ayushman Bharat Digital Health Mission**. **Hence pair 2 is correctly matched.**
 - Chips to Startup (C2S) Programme: The C2S Programme aims to train specialized manpower in the area of design of semiconductor chips at Bachelors, Masters and Research levels, and act as a catalyst for the growth of Startups involved in semiconductor design in the country.

Q 56.B

- Biodiversity can be defined as a community of all the living organisms on the earth and the diversity among them from all the ecosystems. Biodiversity is thus the variability between the species, within the species, and between the ecosystem. Biodiversity is the term popularized by sociobiologist Edward Wilson to describe the combined diversity at all levels of biological organization. The most important of them are–
 - **Genetic diversity:** A single species might show high diversity at the genetic level over its distributional range. The genetic variation shown by the medicinal plant *Rauwolfia vomitoria* growing in different Himalayan ranges might be in terms of the potency and concentration of the active

chemical (reserpine) that the plant produces. India has more than 50,000 genetically different strains of rice, and 1,000 varieties of mango.

- **Species diversity:** The diversity at the species level, for example, the **Western Ghats have a greater amphibian species diversity than the Eastern Ghats. Hence, statement 2 is correct.**
- **Ecological diversity:** At the ecosystem level, India, for instance, with its deserts, rain forests, mangroves, coral reefs, wetlands, estuaries, and alpine meadows have a greater ecosystem diversity than a Scandinavian country like Norway.
- **Patterns of Biodiversity**
 - The diversity of plants and animals is not uniform throughout the world but shows a rather uneven distribution. For many groups of animals or plants, there are interesting patterns in diversity, the most well-known being the latitudinal gradient in diversity.
 - **In general, species diversity decreases as we move away from the equator toward the poles.** With very few exceptions, tropics (latitudinal range of 23.5° N to 23.5° S) harbor more species than temperate or polar areas. Colombia located near the equator has nearly 1,400 species of birds while New York at 41° N has 105 species and Greenland at 71° N only 56 species. **Hence, statement 1 is not correct.**
 - India, with much of its land area in the tropical latitudes, has more than 1,200 species of birds. A forest in a tropical region like Ecuador has up to 10 times as many species of vascular plants as a forest of equal area in a temperate region like the Midwest of the USA.
 - The largely tropical Amazonian rain forest in South America has the greatest biodiversity on earth- it is home to more than 40,000 species of plants, 3,000 of fishes, 1,300 birds, 427 mammals, 427 amphibians, 378 reptiles, and of more than 1,25,000 invertebrates.

Q 57.D

- **The Environment (Protection) Act** was enacted in 1986 with the objective of providing for the protection and improvement of the environment. It empowers the Central Government to establish authorities charged with the mandate of preventing environmental pollution in all its forms and to tackle specific environmental problems that are peculiar to different parts of the country.
- **The Environment (Protection) Rules, 1986** lay down procedures for setting standards of emission or discharge of environmental pollutants.
- It gives certain rights to the **Central Government** which include the following:
 - **Set standards for the emission of pollutants** from any source.
 - Laying down procedures and safeguards for **handling hazardous substances.**
 - Making **rules in matters pertaining to environmental protection as well as to pollution.**
- **Hence, option (d) is the correct answer.**

Q 58.A

- **Recent context: The first I2U2 (India, Israel, United States, and the United Arab Emirates) leaders' Summit was held virtually.**
- **I2U2 stands for India, Israel, the UAE, and the US, and was also referred to as the 'West Asian Quad'.** I2U2 was initially formed in October 2021 following the Abraham Accords, to deal with issues concerning maritime security, infrastructure, and transport. (The Abraham Accord is the first Arab-Israeli peace deal in 26 years mediated by the USA.)
- During the summit, The United Arab Emirates (UAE) announced an investment of USD 2 billion in India to develop Food Parks across the country. The Grouping declared to support a "hybrid renewable energy project" in Gujarat, consisting of 300 megawatts (MW) of wind and solar capacity.
- Six areas of cooperation have been identified by the countries mutually, and the aim is to encourage joint investments in water, energy, transportation, space, health, and food security.
- **Hence option (a) is the correct answer.**

Q 59.B

- **The famous Jagannath Puri Rath Yatra was held in Odisha.** Believed to be the oldest chariot procession in the world, this festival is unique where three Hindu gods are taken out of their temples in a colorful procession to meet their devotees. Lord Jagannath is a form of Hindu gods Krishna, Vishnu, and Rama worshipped in Odisha. Jagannath means 'Lord of the world'.
- The sacred Jagannath Temple is located in Puri, Odisha, and is one of the Char Dham pilgrimage sites for the Hindus. The temple is sacred to all Hindus, especially in those of the Vaishnava traditions. The biggest chariot procession takes place in Puri in the eastern state of Odisha. It is celebrated particularly in Odisha, Jharkhand, and parts of Eastern India.

- **The Jagannath Puri Rath Yatra marks the annual ceremonial procession of Lord Jagannath, his elder brother Balabhadra and younger sister Subhadra, from their home temple to another temple, located in what is believed to be their aunt's home.** This is documented in undated Hindu sacred texts, the Puranas which are believed to have been written a few thousand years ago.
- **Rath Yatra or the Car festival is one of the main festivals of Puri, celebrated every year on Ashad.** Shukla Dwitiya generally falls between the months of June-July. **Hence option (d) is correct.**
- The unique feature of the temple is that Krishna is worshipped not with a spouse but with his siblings, his elder brother Balabhadra and his younger sister Subhadra accompanied by the Sudarshana Chakra.
- **The images are malformed, with no hands or feet, and disproportionately large heads. They are not made of metal or stone, but of wood and cloth and resin and have been replaced from time to time. Hence option (b) is not correct.**
- The story goes that when the image was being carved, the artisan had asked the patron king not to open the door until the work on the idols was completed. But the impatient king did open the door of the workshop, as he did not hear sounds of the wood being carved and polished. Thus, the idol was left incomplete.
- During the festival, the deities are decorated with over 208 kg of gold to complete their remaining limbs.
- **Goti-pua, literally meaning 'single boy', is a temple dancing tradition kept alive by young male dancers, and has heavily influenced the modern-day classical dance, Odissi.**
- After the decline of mahari/devadasi (female temple dancers) system in the sixteenth century due to the diminishing of their social and financial status, the Gotipua dance—performed by a group of pubescence boys impersonating female expressions—became the preferred divine performance in the temple of Lord Jagannath.
- **During the Rath Yatra, the Gotipua performance starts when the deities are taken to their respective chariots in a pompous procession called Pahandi, which means coming step by step or with gradual leaps.** It continues, along with the other rituals and performance tributes, till the end.
 - After the decline of mahari/devadasi (female temple dancers) system in the sixteenth century due to the diminishing of their social and financial status, the Gotipua dance—performed by a group of pubescence boys impersonating female expressions—became the preferred divine performance in the temple of Lord Jagannath. **Hence, option (c) is correct.**
 - During the Rath Yatra, the Gotipua performance starts when the deities are taken to their respective chariots in a pompous procession called Pahandi, which means coming step by step or with gradual leaps. It continues, along with the other rituals and performance tributes, till the end.
- **UNESCO World Intangible Cultural Heritage List : All Inclusions From India:**
 - Kutiyattam, Sanskrit theatre
 - The tradition of Vedic chanting
 - Ramlila, the traditional performance of the Ramayana
 - Ramman, religious festival and ritual theatre of the Garhwal Himalayas, India
 - Chhau dance
 - Kalbelia folk songs and dances of Rajasthan
 - Mudi yettu, ritual theatre and dance drama of Kerala
 - Durga Puja in Kolkata
 - Buddhist chanting of Ladakh: recitation of sacred Buddhist texts in the trans-Himalayan Ladakh region, Jammu and Kashmir, India
 - Sankirtana, ritual singing, drumming and dancing of Manipur
 - Traditional brass and copper craft of utensil making among the Thatheras of Jandiala Guru, Punjab, India
 - Nawrouz, Novruz, Nowrouz, Nowrouz, Nawrouz, Nauryz, Nooruz, Nowruz, Navruz, Nevruz, Nowruz, Navruz
 - Yoga
 - Kumbh Mela
- **Jagannath Rath Yatra is not included in the UNESCO World Intangible Cultural Heritage List. Hence option (a) is correct.**

Q 60.C

- **Green House Gases (GHGs):** Gases that help in causing greenhouse effect are called greenhouse gases (GHGs). These gases either occur naturally or are produced on earth due to human activities of burning fossil fuel and biomass. One of the most abundant naturally occurring greenhouse gases is water vapour. Other greenhouse gases are carbon dioxide, Methane, Nitrous oxide, Trifluoromethyl sulphur pentafluoride and hydrochlorofluorocarbons. It is since the 1700s, that a substantial increase in the concentration of greenhouse gases has occurred in the atmosphere.

- **Water Vapour:** It accounts for about 60 to 70 per cent of the natural greenhouse effect. Its level in the atmosphere rises with the increasing global warming adding up further to the greenhouse effect. **Hence, statement 1 is not correct.**
- **Carbon dioxide:** In the modern age of industrialization and increasing automobile exhausts the concentration of carbon dioxide is increasing faster than the earth's natural capacity of assimilation. Currently, the CO₂ concentration in the atmosphere is about 370 parts per million (ppm). It accounts for more than 60 per cent of the additional greenhouse effect.
- **Methane:** This gas is produced through various sources like **decomposing organic substances, coal mining, paddy fields, production and transport of other fossil fuels etc.** Its concentration in the atmosphere has become more than double since 1750. Scientists are of the opinion that it is an extremely effective heat trapping gas. One molecule of methane is 20 times more efficient in terms of trapping infrared radiation than a molecule of carbon dioxide. **Hence, statement 3 is correct.**
- **Nitrous Oxide:** This gas is released into the atmosphere by burning of fossil fuels, automobile exhaust, decomposition of nitrogenous fertilizers in the soil etc. This gas **has a capacity of trapping heat 300 times more effectively than carbon dioxide.** It can stay in the atmosphere for about 100 years. **Hence, statement 2 is correct.**
- **Fluorinated Compounds:** Compounds comprising CFCs (chlorofluoro-carbons), HCFCs (hydrochlorofluorocarbons) and HFCs (hydrofluorocarbons) are man-made compounds called as fluorinated compounds. These compounds are used in a variety of manufacturing processes. Each molecule of these synthetic compounds is many thousand times more effective in trapping infrared radiations than a single molecule of carbon dioxide.
- **Trifluoromethyl sulphur pentafluoride:** Each molecule of this industrially produced compound **can trap heat more effectively than all the other gases known to cause green house effect.**

Q 61.A

- **Use of medicinal herbs in the treatment of diseases:** The term "medicinal plant" include various types of plants used in herbalism ("herbology" or "herbal medicine"). It is the use of plants for medicinal purposes and the study of such uses. Medicinal plants play vital roles in disease prevention. Some of them are:
 - **Natakarn tree:** The leave of natakanar tree is used to cure stomach disorders in cattle. The tuber from its saplings is given to the cattle to increase their appetite.
 - **Gondvil:** Thorny creepers such as the Gondvil are used to cure mouth ulcers in cattle.
 - **Karnavand:** It has medicinal properties. It is commonly found in Kolhapur and Konkan belts. Its roots and flowers are used to treat stomach disorders, scabies, skin diseases, and fever. **Hence pair 1 is correctly matched.**
 - **Sitecha Ashok:** The sitecha ashok is commonly used for pimples and other skin ailments. **Hence pair 2 is correctly matched.**
 - **Katal jasvand:** It is a medicinal plant found in the sacred grove. Its tuber is used to cure joint pains. **Hence pair 3 is correctly matched.**
- **Examples of some more herbs with their medicinal values**
 - **Lemongrass:** The aromatic plant, lemongrass is a rich source of vitamin B₂, vitamin B₁, vitamin A, vitamin C, folate, potassium, manganese, and other essential minerals. It not only makes your tea or other dishes flavorful but also has many health benefits. It relieves menstrual pain and sore throat. It can also be used to treat sleep trouble and stress. The aromatic herb also has antipyretic properties and thus can be used for the treatment of fever.
 - **Methi:** Methi leaves are a popular vegetable in India. Both the leaves and the seeds have several health benefits. Fenugreek leaves keep our cholesterol in check, treat respiratory problems, cure fever, reduce blemishes, give you thick, long hair, cure anemia and treat digestive issues.
 - **Pudina:** Pudina not only adds a refreshing taste and delicious aroma to a dish or drink but also has several medicinal properties. It can be used for curing indigestion. It also helps in healing bruises. A Mint face pack can help you get rid of acne and blackheads. You can use it as a skin toner and strengthen your hair. Use a concoction made of mint and celery to rinse your hair. This will strengthen your hair.
 - **Marigold:** The fragrant flowers can also be used for the treatment of skin issues. Marigold is an excellent remedy for blemishes, acne, and sunburn. Ointment or lotion with marigold extract can treat skin issues. It can also be used for treating digestive problems and ulcers. All you need to do is make marigold tea to get relief.
 - **Holy basil:** Apart from religious beliefs, **Tulasi** leaves are rich in antioxidants and also exhibit anti-fungal properties. The antioxidant-rich species is found to be widely used in herbal teas, beauty

products, and also medicines to fight Indigestion, diabetes, fever, cold, cough, and bronchitis. Apart from its medicinal properties, tulsi also cleans the air and is one of the air-purifying plant-list by NASA.

- **Moringa:** Moringa oleifera is a small fast – growing evergreen or deciduous tree that usually grows up to 10 or 12 m in height. It's a tropical tree that can survive droughts. Moringa is often called the drumstick tree because of its skinny, foot-long pods. It also goes by mother's best friend, the miracle tree, the never die tree and the ben oil tree. Moringa's peppery leaves are often eaten as a vegetable. They're also dried and ground into a powder used in soups and curries. They have iron, potassium, and calcium. Moringa bark extract has been used to help treat stomach issues, anemia, diabetes, and other conditions. Moringa is a distant cousin of broccoli, kale, and cabbage. The roots can be ground to make a paste that tastes like horseradish. The paste can help with snakebites, toothaches, and malaria.
- **Spearmint:**
 - Spearmint is known by many names like Mackerel mint, Garden mint, Lamb mint, and common mint. The leaves of these plants are rich in Vitamins, Anti-oxidants, and Manganese and are popularly used in Summer drinks, teas, and chutneys across India.
 - The Aroma of these mint leaves is enough to provide one with a refreshing feel and is also proven to keep mosquitoes away in the winter. Mint leaves help to refrain sore throat, cramps, Arthritis, Diarrhea, Fatigue, and flatulence. Mint essential oils are also used in relieving muscle pains and are quite often used in ayurvedic spas for soothing stress-free body massage.
- **Carom:**
 - Carom seeds also known as Ajwain are one of the most commonly found home and kitchen ingredients. These seeds are sprinkled or added to almost every Indian curry recipe. Ajwain seeds do possess a strong Aroma which not only adds up to the flavor of the food but also adds up to many health benefits.
 - Carom seeds are rich in Antioxidants, Vitamins, minerals, and fiber and also exhibit antibacterial and antifungal properties. Carom seeds do help in relieving many common human ailments like Indigestion, Ulcers, Acidity, Arthritis, Bad Cholesterol, Blood pressure, and even common cold or cough.
- **Giloy:**
 - Gilroy, a herb that is termed as the root of Ayurvedic immortality was being used in Indian Medicines for ages due to its highly rich medicinal properties. Giloy is quite famous and commonly found in Tropical regions of the country and also it's quite simple and easy to grow in one's home just as you do with a Money plant.
 - Unlike other species, Giloy herbs are particularly famous for their ability to increase platelet count and it's really helpful for people suffering from low blood platelets. Apart from that, these herbs help solve Diabetes, Arthritis, Dengue, Chikungunya, and Indigestion, combat respiratory problems, and also maintain healthy heart conditions
 - Chewing on Giloy roots can help asthma patients relieve breathing problems. Boiling water with Giloy leaves or Giloy powder will help boost Immunity and Memory power. People usually consume Giloy juice as a detox drink as it is believed to act as an Anti-aging and reduces sprinkles, pores, and wrinkles, and reward a fair skin tone.
- **Aloe Vera:**
 - One such Ayurvedic shrub which doesn't require much introduction, Aloe Vera is a quite commonly found stemmy shrub in every household and neighborhood. Aloe gel which is used across many medicines and a lot of beauty and facial products is derived from the leaves of the Aloe Vera shrub. This species has been in the market for decades and is now a very popular shrub above all.
 - Aloe Vera does exhibit some antioxidant and antimicrobial properties and also helps in combating Constipation, Ulcers, and Irritative Lower Bowel Syndrome (IBS) and also helps maintain blood sugar levels. Aloe gel is quite commonly used in Tooth gels, Mouthwashes, and many food flavoring agents.
- **Curry Leaves:**
 - Curry Leaves also known as Kadai Patta is every kitchens delicacy. The leaves of Curry trees are used as seasoning and flavoring agents in almost all Indian foods and literally found in all tropical and subtropical regions of the country. Every part of the plant: leaves, roots, and bark are quite largely used for various medicinal applications. But Curry leaves are quite famous due to their high nutritional benefits and also their use as a flavoring ingredient.

- Curry leaves are rich in minerals, and fibers and also help in boosting immunity. It helps in relieving Aemia. Diabetes, Indigestion problems, cholesterol, and helps maintain healthy liver conditions.
- **Khus:**
 - Khus plants or popularly known as Khus grass are widely found in plains and hilly tropical areas of India. Apart from Medicinal applications, the leaves are also used for the manufacturing of Mats, and hand fans, and also serve many chemical applications.
 - The roots of these plants possess a high-level fragrance and it is the most important part of this plant. Oils made from Khus roots are used in high-end perfumes and high-end soaps due to their fragrance. The Khus plants have high medicinal properties and help prevent Jaundice, Diuretic, and Arthritis, Maintains Blood sugar levels, and also boost blood circulation and immunity levels.
- **Ashwagandha:**
 - Ashwagandha is a highly lucrative medicinal herb whose application comes again from 3000 years ago and is a vital ayurvedic herb assisting natural healing. The root and berry(fruit) are the most important parts of the plant and their extracts are used in a wide range of medicinal applications.
 - Ashwagandha extracts are mainly implied as an adaptogen to relieve stress and depression. Further, these extracts also help in curing ailments like Arthritis, Bipolar Disorder, Hyperactivity Disorder, OCD, and Insomnia and also help maintain Blood sugar levels, Cortisol levels, and cholesterol levels.

Q 62.B

- Bioprospecting is defined as a systematic and organized search for useful products derived from bioresources including plants, microorganisms, animals, etc., that can be developed further for commercialization and overall benefits of the society.
- **It means exploring molecular, genetic and species-level diversity for products of economic importance.**
- Bioprospecting activities must comply with the definition of the utilization of genetic resources of the **Nagoya Protocol** or as stated in the national law or policy.
 - The **Nagoya Protocol applies to the utilization of genetic resources and their derivatives**. The rationale is to extract the maximum commercial value from genetic resources and indigenous knowledge, while creating a fair compensation system that can benefit all. The phases of bioprospecting start with sample collection, isolation, characterization and move to product development and commercialization. Bioprospecting is possible both in terrestrial and marine environments. Many molecules, such as **trabectedin (an antitumor agent) and eribulin (used to treat breast cancer)**, were discovered from marine organisms.
- Bioprospecting, when properly regulated, generates revenues that can be directly linked to the conservation of biodiversity and to the benefit of local communities.
- **Hence, option (b) is the correct answer.**

Q 63.B

- **Recent context: Ending uncertainties over crucial appointments to state-run banks, general insurers, and other financial institutions, the government recently decided to set up the Financial Services Institution Bureau (FSIB) with a wider mandate to replace the Banks Board Bureau (BBB).** The government was forced to replace the BBB with a new entity after the Delhi High Court had in the last year 2021 ruled that the BBB couldn't select the general managers and directors of state-run general insurers, as it was not a competent body.
- The functions of FSIB shall be as mentioned as follows, namely:-
 - **To recommend persons for appointment as whole-time directors (WTDs) and nonexecutive chairpersons (NECs) on the Boards of Directors in Public Sector Banks, financial institutions, and Public Sector insurers** (hereinafter referred to as "PSBs", "FIs" and "PSIs" respectively); **Hence statement 1 is correct.**
 - To advise the government on matters relating to appointments, transfer or extension of the term of office, and termination of services of the said directors;
 - To advise the government on the desired management structure at the Board level for PSBs, FIs, and PSIs;
 - To advise the Government on a suitable performance appraisal system for WTDs and NECs in PSBs, FIs, and PSIs;
 - To build a databank containing data related to the performance of PSBs, FIs, and PSIs.

- The Chairperson of FSIB is to be nominated by the Central Government, who shall be a retired official banking sector or a regulatory institution, or a business person of repute with sufficient knowledge of the financial sector, or a person with at least 25 years of experience in public administration with experience in banking and the financial sector. Hence statement 2 is not correct.
- After the formation of FSIB, all assets, interests, and liabilities of the Banks Board Bureau stand transferred to FSIB; all books of accounts, registers, records, and all other documents of the Banks Board Bureau stand transferred to FSIB; any cause of action, suit, decree, appeal or other proceeding pending by or against Banks Board Bureau before any court or other authority may be continued and enforced by or against FSIB. Hence statement 3 is correct.

Q 64.B

- **Tadoba National Park:** Located in Chandra Pur district of Maharashtra state of India, this National Park was notified in the year 1955. It is spread in an area of 11, 655 hectares and it supports the populations of tiger, sambhar, sloth bear, lion, chital, chinkara, barking deer, blue bull, four horned deer, langur, peafowl and crocodile.
- **Bandhav Garh National Park:** This Park is located in Madhya Pradesh. The principal animal kept in this park is the White Tiger. Notified in the year 1968, this national park is spread in the area of 44,884 hectares.
- **Dudwa National Park:** Located in Lakhimpur Khiri district of Uttar Pradesh, this National Park supports wide variety of wild animals including re-introduced one-horned rhinos and swamp deer etc. Other animals in this park are – crocodiles, leopards, jackals, sambhars and sloth bears. Principal plants comprise grass species, Sal trees etc. This national park was notified in the year 1977 and its area is 49, 029 hectares.
- Hence, option (b) is the correct answer.

Q 65.D

- **Ozone layer depletion:** It refers to the thinning of the protective ozone layer in the atmosphere. This happens when certain chemicals come into contact with ozone and destroy it. Chemical compounds that cause ozone layer depletion are called Ozone Depleting Substances (ODS).
- **Some of the substances that cause depletion of the Ozone Layer are:**
 - **Chlorofluorocarbons, methane, nitrous oxides (N₂O), carbon tetrachloride (CCl₄), methyl bromide (a soil fumigant and insecticide), aircraft emissions, n-propyl bromide, and Halon-1202** are major agents that cause depletion of the ozone layer. Hence, these are called Ozone Depleting Substances (ODS).
 - **Chlorofluorocarbons are a group of aliphatic organic compounds.** These are a family of synthetic chemicals that are mostly the compounds of chlorine, fluorine, and carbon. These are stable, nonflammable, non-corrosive chemicals with a peculiar trade name Freon. Being relatively non-toxic chemicals, these are easy and inexpensive to produce.
 - During the 1970s CFCs were linked to the destruction of the ozone layer due to which its manufacture has been banned in most countries of the world. Some important members of CFC group are **dichlorodifluoromethane (Freon-12), trichlorofluoromethane (Freon-11), chlorodifluoromethane (Freon-22), dichlorotetrafluoroethane (Freon-114) and trichlorotrifluoroethane (Freon-113).** On earth, these chemicals are used extensively as aerosol spray propellants, refrigerants, solvents, and foam blowing agents.
 - Other ODS include hydrochlorofluorocarbons (HCFCs), halons, hydrobromofluorocarbons, chlorobromomethane, and methyl chloroform. ODS are generally very stable in the troposphere and only degrade under intense ultraviolet light in the stratosphere.
 - Hence option (d) is the correct answer.

Q 66.B

- The Central Government appoints a Director of Wild Life Preservation. Hence, statement 1 is not correct.
 - According to the Wildlife Protection Act, 1972, the Director of Wild Life Preservation who shall be the Member-Secretary of the National Board of Wildlife.
- It prohibits hunting of endangered species. **It bans trade and commerce in scheduled animals.** There is a provision for trade in some wildlife species with a license for sale, possession, transfer etc. Hence, statement 2 is correct.
- The specified endemic plants in Schedule VI are prohibited from cultivation and planting. **The Schedule VI of the Act, provides protection to 6 plants: Hence, statement 3 is not correct.**

- Beddome's cycad (*Cycas beddomei*)
- Blue Vanda (*Vanda soerulec*)
- Kuth (*Saussurea lappa*)
- Ladies slipper orchids (*Paphiopedilum* spp.)
- Pitcher plant (*Nepenthes khasiana*)
- Red Vanda (*Ranantthera imschootiana*)

Q 67.D

- **Water pollution:** The most serious water pollutants are the disease causing agents called pathogens. Pathogens include bacteria and other organisms that enter the water from domestic sewage and animal excreta. Human excreta contain bacteria such as *Escherichia coli* and *Streptococcus faecalis* which cause gastrointestinal diseases.
- **Organic chemicals** are another group of substances that are found in polluted water. Petroleum products pollute many sources of water e.g., major oil spills in oceans. Other organic substances with serious impacts are the pesticides that drift down from sprays or runoff from lands. Various industrial chemicals like polychlorinated biphenyls, (PCBs) which are used as cleaning solvents, detergents, and fertilizers add to the list of water pollutants. PCBs are suspected to be carcinogenic.
- Nowadays most of the **detergents** available are biodegradable. However, their use can create other problems. The bacteria are responsible for degrading biodegradable detergent feed on it and grow rapidly. While growing, they may use up all the oxygen dissolved in water. The lack of oxygen kills all other forms of aquatic life such as fish and plants.
- **Fertilizers** contain phosphates as additives. The addition of phosphates in water enhances algae growth. Such profuse growth of algae covers the water surface and reduces the oxygen concentration in water. This leads to anaerobic conditions, commonly with an accumulation of obnoxious decay and animal death. Thus, bloom-infested water inhibits the growth of other living organisms in the water body. This process in which nutrient-enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity is known as Eutrophication.
- **Fluoride:** For drinking purposes, water should be tested for fluoride ion concentration. Its deficiency in drinking water is harmful to man and **causes diseases such as tooth decay** etc. Soluble fluoride is often added to drinking water to bring its concentration up to 1 ppm or 1 mg/dm³. The F⁻ ions make the enamel on teeth much harder by converting hydroxyapatite, the enamel on the surface of the teeth, into much harder fluorapatite. **Hence pair 2 is not correctly matched.**
- **Lead:** Drinking water gets contaminated with lead when lead pipes are used for the transportation of water. The prescribed upper limit concentration of lead in drinking water is about 50 ppb. **Lead can damage kidneys, liver, reproductive system, etc. Minamata disease is a neurological disease caused by severe mercury poisoning. Hence pair 1 is not correctly matched.**
- **Sulfate:** Excessive sulfate (>500 ppm) in drinking water causes a laxative effect, otherwise at moderate levels it is harmless.
- **Nitrate:** The maximum limit of nitrate in drinking water is 50 ppm. **Excess nitrate in drinking water can cause diseases such as methemoglobinemia ('blue baby' syndrome). Hence pair 3 is not correctly matched.**

Q 68.A

- **ECOLOGICAL SUCCESSION**
 - The gradual and fairly predictable change in the species composition of a given area is called **ecological succession**.
 - An important characteristic of all communities is that their composition and structure constantly change in response to the changing environmental conditions. This change is orderly and sequential, parallel with the changes in the physical environment. These changes lead finally to a community that is in near equilibrium with the environment and that is called a **climax community**.
 - During succession, some species colonize an area and their population becomes more numerous whereas populations of other species decline and even disappear. The **entire sequence of communities that successively change in a given area** is called **sere(s)**. **Hence statement 1 is correct.**
 - The individual transitional communities are termed seral stages or seral communities. In the successive seral stages, there is a change in the diversity of species of organisms, an increase in the number of species and organisms as well as an increase in the total biomass.
 - Succession is a process that starts in an area where no living organisms are there – these could be areas where no living organisms ever existed, say bare rock; or in areas that somehow, lost all the

living organisms that existed there. The former is called **primary succession**, while the latter is termed **secondary succession**.

- **Examples of areas where primary succession occurs** are newly cooled lava, bare rock, and a newly created pond or reservoir. The establishment of a new biotic community is generally slow. Before a biotic community of diverse organisms can become established, there must be soil. Depending mostly on the climate, it takes natural processes several hundred to several thousand years to produce fertile soil on bare rock.
- **Secondary succession** begins in areas where natural biotic communities have been destroyed such as in abandoned farmlands,
- burned or cut forests, and lands that have been flooded. Since some soil or sediment is present, succession is faster than primary succession.
- Description of ecological succession usually focuses on changes in vegetation. However, these vegetational changes, in turn, affect food and shelter for various types of animals. Thus, as succession proceeds, the numbers and types of animals and decomposers also change.
- **Succession of Plants**
 - Based on the nature of the habitat – whether it is water (or very wet areas) or it is on very dry areas – a succession of plants is called hydrarch or xerarch, respectively. **Hydrarch succession takes place in wet areas** and the successional series progress from hydric to mesic conditions.
 - As against this, **xerarch succession takes place in dry areas**, and the series progresses from xeric to mesic conditions. Hence, both hydrarch and xerarch successions lead to **medium water conditions (mesic) – neither too dry (xeric) nor too wet (hydric)**. **Hence statement 3 is not correct.**
 - The species that invade a bare area are called pioneer species. In primary succession on rocks, these are usually lichens which are able to secrete acids to dissolve rock, helping in weathering and soil formation.
 - These later pave the way for some very small plants like bryophytes, which are able to take hold in the small amount of soil. They are, with time, succeeded by higher plants, and after several more stages, ultimately a stable climax forest community is formed. The climax community remains stable as long as the environment remains unchanged. With time the xerophytic habitat gets converted into a mesophytic one.
 - **In primary succession in water**, the pioneers are the small phytoplankton, which are replaced with time by rooted-submerged plants, rooted-floating angiosperms followed by free-floating plants, then reed swamp, marsh-meadow, scrub, and finally the trees. The climax again would be a forest. With time the water body is converted into the land.
 - **In secondary succession** the species that invade depend on the condition of the soil, availability of water, the environment, and also the seeds or other propagules present. Since soil is already there, the **rate of succession is much faster** and hence, the climax is also reached more quickly. **Hence, statement 2 is correct.**
 - What is important to understand is that succession, **particularly primary succession, is a very slow process**, taking maybe thousands of years for the climax to be reached.
 - Another important fact is to understand that **all succession whether taking place in water or on land, proceeds to a similar climax community – the mesic.**

Q 69.C

- Captive breeding programs breed endangered species in zoos and other facilities to build a healthy population of the animals and, sometimes, to reintroduce endangered species back into the wild.
- **It can be done in zoos, aquariums, National Parks etc. Hence, statement 2 is not correct.**
- **It provides a means for conserving species that may not survive in the wild. Hence, statement 1 is correct.**
- Captive populations are maintained for many reasons like conservation, education, and exhibition of interesting species. Establishing captive populations for saving species from extinction is an important contribution of zoos to conservation.
- **Nandankanan in Odisha is the first-ever captive breeding centre for creating awareness of wildlife in India. Hence, statement 3 is correct.**
- Unlike other zoos in the country, Nandankanan is built right inside the forest and set in a completely natural environment. Nandankanan is the first zoo in the world to breed white tigers and melanistic tigers.

Q 70.C

- The **International Swimming Federation (FINA)** voted recently approved a new gender inclusion policy to restrict transgender athletes to participate in elite female aquatics and swimming competitions. The transgender athlete who have completed their transition before the age of 12 or before they reach stage 2 of the puberty Tanner Scale shall be allowed to compete in the women's category.
- **Tanner Scale:** The Tanner scale, developed by British pediatrician James Tanner, **provides a rating system for physical development in children and adolescents. It describes the onset and progression of pubertal changes.** The scale defines physical development based on external primary and secondary sex characteristics.
- **Hence option (c) is the correct answer.**

Q 71.C

- **Coral reefs:** Corals are massive wave-resistant structures built largely by coral and consisting of skeletal and chemically precipitated material. These are made by polyps i.e. animals belonging to the class Coelenterata.
 - In the Indian seas, coral reefs **are found in the Gulf of Mannar, Palk-bay, Lakshadweep (atolls), Gulf of Kutch, and Andaman and Nicobar sea areas. Hence option (c) is the correct answer.**
 - The health of coral reefs has been declining over the past several decades. The pollution of seawater, destructive fishing practices, algal blooms, eutrophication, coral diseases, and coral bleaching are some serious causes of the depletion of coral reefs.
- **What is Coral Bleaching?**
 - The whitening of coral colonies due to the loss of symbiotic zooxanthellae from the tissues of polyps is called Coral Bleaching.
 - Zooxanthellae are unicellular algae that provide color to corals. They also provide food to corals and their deaths lead to the breakdown of the symbiotic relationship between them. This breakdown of symbiotic relationships causes starvation and resultant deaths of coral polyps. This condition exposes the white calcium carbonate skeletons of the coral colony.
 - There are a number of stresses or environmental changes that may cause bleaching. These causes include disease, excess shade, increased levels of ultraviolet radiation, sedimentation, pollution, salinity changes, and increased temperatures.
- **Other causes of coral bleaching include -**
 - Increased exposure to ultraviolet (UV) radiation;
 - Large amounts of stormwater from heavy rains flooding the reef;
 - The exposure of coral to certain chemicals or diseases;
 - Sediments such as sand or dirt covering the coral;
 - Excess nutrients such as ammonia and nitrate from fertilizers and household products enter the reef ecosystem.
 - (The nutrients might increase the number of zooxanthellae in the coral, but it is possible that the nutrient overload increases the susceptibility of coral to diseases.)

Q 72.C

- **Why in news ?**
 - Health Minister emphasized for global recognition for Indian Pharmacopoeia:
 - India has become 'pharmacy of the world' by specialising in generic medicine formulation.
 - Pharmacopoeia is important to develop a Swasthya and Samrudh Bharat, to maintain standard quality of our medical products — vaccines, medicines, equipment etc. and to keep an eye on the effect of these medicines on patients, said Union Health Minister.
- The Govt. of India has created a separate, dedicated, autonomous institution in the form of the Indian Pharmacopoeia Commission (IPC) to deal with matters relating to the timely publication of the Indian Pharmacopoeia which is the official book of standards for drugs included therein, in terms of the Second Schedule to the Drugs and Cosmetics Act, 1940 so as to specify the standards of identity, purity and strength of the drugs imported, manufactured for sale, stocked or exhibited for sale or distributed in India.
- The mandate of the Commission is to perform, inter-alia, functions such as revision and publication of the Indian Pharmacopoeia and National formulary of India on a regular basis besides providing IP Reference Substances and training to the stakeholders on Pharmacopoeial issues. **The Commission has become fully operational from 1st January 2009 as an autonomous Body, fully financed by the Central Government with specific budgetary allocations under the administrative control of the Ministry of Health and Family Welfare. Hence statement 1 is correct.**

- **Mission**
 - To promote public and animal health in India by bringing out authoritative and officially accepted standards for quality of drugs including active pharmaceutical ingredients, excipients and dosage forms, used by health professionals, patients and consumers.
- **Vision**
 - **To promote the highest standards of drugs for use in humans and animals within practical limits of the technologies available for manufacture and analysis. Hence statement 2 is correct.**
- **Objectives**
 - To develop comprehensive monographs for drugs to be included in the Indian Pharmacopoeia, including active pharmaceutical ingredients, pharmaceutical aids and dosage forms as well as medical devices and to keep them updated by revision on a regular basis.
 - **To develop monographs for herbal drugs, both raw drugs and extracts/formulations therefrom.**
 - To accord priority to monographs of drugs included in the National Essential Medicines List and their dosage forms.
 - To take note of the different levels of sophistication in analytical testing/ instrumentation available while framing the monographs.
 - To accelerate the process of preparation, certification and distribution of IP Reference Substances, including the related substances, impurities and degradation products.
 - To collaborate with pharmacopoeias like the Ph Eur, BP, USP, JP, ChP and International Pharmacopoeia with a view to harmonizing with global standards.
 - To review existing monographs periodically with a view to deleting obsolete ones and amending those requiring upgrading /revision.
 - To organize educational programs and research activities for spreading and establishing awareness on the need and scope of quality standards for drugs and related articles /materials.
 - To publish the National Formulary of India for updating medical practitioners and other healthcare professionals.
 - To act as a National Coordination Centre for Pharmacovigilance Programme of India.

Q 73.C

- **Effects of Pollution on Plants and Microorganisms:** Most types of pollution are harmful to plants and microorganisms as they are harmful to humans and animals. Some remarkable examples are mentioned below.
 - The high concentrations of sulfur dioxide (SO_2) produce necrosis (death of tissues), blotching of broad-leaved plants and grasses, brownish discolorations of tips of pine needles, chlorosis (gradual yellowing), etc.
 - **Hydrogen Sulphide (H_2S)** in the air causes leaf lesions or spots, defoliation, and reduced growth. **Hence statement 1 is correct.**
 - **High concentration (100 to 10,000 parts per million) of carbon monoxide causes leaf – drop, reduction in leaf size, premature aging, inhibition of cellular respiration, etc. Hence statement 2 is correct.**
 - Hydrogen fluoride causes burning of tips of leaves, excessive dropping of blooms and fruits, development of seedless and small fruits, etc.
 - The concentration of Hydrogen chloride in the air is reported to be responsible for the abaxial glazing of leaves caused by the collapse and plasmolysis of epidermal cells.
 - Normal to the high concentration of hydrocarbons like ethylene causes dry sepal disease of Orchids, decrease in the amounts of chlorophyll and carotenoids, shortening of internodes, and lack of apical dominance.
 - Ammonia, nitrogen oxides, and fluorides cause different types of harmful effects on leaves, stems, and fruits of plants. Nitrogen oxides, ozone, and PAN are reported to disturb the metabolic processes of plants and cause the death of tissues. (PAN = Perox Acyle Nitrates)
 - The application of pesticides and fertilizers kills a number of soil organisms, worms, and insects. Many useful bacteria are killed in water due to contamination with pesticides. These bacteria are useful for the decomposition of bio-degradable wastes in water leading to its purification.
 - Dust and silica particles from crushers, lime kilns, slate-making units, and quarries cause serious damage to plants. Particles released from cement manufacturing units are responsible for the premature fall of needles, formation of more stomata, reduction in the weight of seeds, and increase in the number of infertile seeds.

Q 74.A

- Ozone gas is continuously formed by the action of Ultra-Violet (UV) rays on molecular oxygen, and also degraded into molecular oxygen in the stratosphere. There should be a balance between production and degradation of ozone in the stratosphere. Of late, the balance has been disrupted due to enhancement of ozone degradation by chlorofluorocarbons (CFCs).
- **UV radiation of wavelengths shorter than UV-B, are almost completely absorbed by Earth's atmosphere, given that the ozone layer is intact. But, UV-B damages DNA and mutation may occur. It causes aging of skin, damage to skin cells and various types of skin cancers. In human eye, cornea absorbs UV-B radiation, and a high dose of UV-B causes inflammation of cornea, called snow-blindness, cataract, etc. Such exposure may permanently damage the cornea.**
- **Hence option (a) is the correct answer.**

Q 75.D

- **Bioindicators are living organisms such as plants, planktons, animals, and microbes, which are utilized to screen the health of the natural ecosystem in the environment.** Living organisms that show the status of environmental quality through their sensitivity, are called bio-indicators.
- They are used for assessing environmental health and biogeographic changes taking place in the environment. Naturally occurring Bioindicators are used to assess the health of the environment and are also an important tool for detecting changes in the environment.
- Living organisms that show the status of environmental quality through their sensitivity, are called bio-indicators. The bio-indicators are very useful in finding out pollution because they help in the identification of the type of pollution. Let us take a few examples-
 - The **presence of Coliform bacteria** indicates water pollution due to fecal matter.
 - The **movement of Catla and Labeo (Rohu) fish species away from their habitat** shows pollution of water due to industrial waste. Similarly, the death of Amphibians and fish in pond water shows Eutrophication.
 - The **abundance of diatoms and Eichhornia in water** indicate pollution due to the mixing of sewage in the water. **Hence option (d) is the correct answer.**
 - The **growth of some algal species like Chara** shows water pollution by organic and industrial wastes.

Q 76.C

- **Adaptation** is an attribute of the organism (morphological, physiological, behavioral) **that enables the organism to survive and reproduce in its habitat.** Many adaptations have evolved over a long evolutionary time and are genetically fixed.
- In the absence of an external source of water, the kangaroo rat in North American deserts is capable of meeting all its water requirements through its internal fat oxidation (in which water is a by-product). It also has the ability to concentrate its urine so that a minimal volume of water is used to remove excretory products.
- Many desert plants have a thick cuticle on their leaf surfaces and have their stomata arranged in deep pits (sunken) to minimize water loss through transpiration. They also have a special photosynthetic pathway (CAM) that enables their stomata to remain closed during the daytime. Some desert plants like Opuntia, have no leaves – they are reduced to spines– and the photosynthetic function is taken over by the flattened stems.
- **Mammals from colder climates generally have shorter ears and limbs to minimize heat loss. This is called Allen's Rule. Hence statement 1 is correct.**
- In the **polar seas, aquatic mammals like seals have a thick layer of fat (blubber)** below their skin that acts as an insulator and reduces the loss of body heat. **Hence statement 2 is correct.**
- In most animals, the metabolic reactions and hence all the physiological functions proceed optimally in a narrow temperature range (in humans, it is 37°C). But there are microbes (archaebacteria) that flourish in hot springs and deep-sea hydrothermal vents where temperatures far exceed 100°C.

Q 77.A

- **Ola Electric recently uncovered India's first indigenously developed lithium-ion cell called "NMC 2170".** NMC 2170 was built in-house. Mass production of the cell will start at its upcoming Gigafactory by 2023.
- **NMC 2170 is a high nickel cylindrical Ola cell. It comprises of Nickel-Manganese-Cobalt (NMC) on the cathode side while graphite & silicon on the anode side.** It makes use of specific chemicals and materials, which helps the cell in packing more energy in a given space. It also improves the overall life

cycle of the lithium-ion cell. The cell has been developed by considering indigenous conditions at the core.

- **Hence option (a) is the correct answer.**

Q 78.A

- Endemism is a term used in biology to talk about the distribution of a taxon limited to a small geographic area and which can therefore be found naturally in this place.
- **In consequence, endemic species are those that live in a limited area**, such as a mountain range, lake or island, among others. Therefore, the ecological aspect of the place and the biological characteristics of the living beings influence this condition.
- As the endemic species has very narrow ecological niche and requirement for physical environment, habitat conservation is integral part of endemic species.
- **A rare species** is a group of organisms that are very uncommon, scarce, or infrequently encountered. This designation may be applied to either a plant or animal taxon, and is distinct from the term endangered or threatened.
- **A flagship species** is a species selected to act as an ambassador, icon or symbol for a defined habitat, issue, campaign or environmental cause.
- Species that arrive first in a newly created environment (such as an island rising out of the sea) are called **pioneer species**, and they, through their interactions with one another, build a rather simple initial biological community.
- **Hence, option (a) is the correct answer.**

Q 79.C

- Biopiracy is defined as the **privatisation and unauthorised use of biological resources** by entities (including corporations, universities and governments) outside a country which has pre-existing knowledge of rare biological resources. **Biopiracy is the theft or usurpation of genetic materials especially plants and other biological materials by the patent process. Hence, statement 1 is correct.**
- Example: use of indigenous knowledge of medicinal plants for patenting by medical companies without recognizing the fact that the knowledge is not new, or invented by the patenter, And thereby the piracy deprives the indigenous community to the rights to commercial exploitation of the technology that they themselves had developed.
- Agencies indulging in biopiracy illegally claim:
 - exclusive commercial rights to plants, animals, organs, micro-organisms and genes.
 - the commercialisation of traditional communities' knowledge on biological resources.
 - patenting of biological resources.
- Biopiracy is not limited to drug development. It also occurs in agricultural and industrial contexts. Indian products such as the neem tree, tamarind, turmeric, and Darjeeling tea have all been patented by foreign firms for different lucrative purposes.
- **The Convention on Biological Diversity (CBD)** is the international legal instrument for "the conservation of biological diversity, the sustainable use of its components and the fair and equitable sharing of the benefits arising out of the utilization of genetic resources" that has been ratified by 196 nations.
- The **Convention on Biological Diversity** gives **sovereign national rights over biological resources. Hence, statement 2 is correct.**

Q 80.B

- **Acid Rain – Acidification**
 - Acid rain refers to any precipitation (rain, fog, mist, snow) that is more acidic than normal (pH of less than 5.6. pH below 7 is acidic).
 - Acid rain is caused by atmospheric pollution from acidic gases such as sulphur dioxide and oxides of nitrogen emitted from the burning of fossil fuels.
 - It is also recognized that acidic smog, fog, and mist, move out of the atmosphere and settle on dust particles which in turn accumulate on vegetation as acid depositions.
 - When rain falls, the acid from these depositions leaks and forms acid dews.
- **‘Spring shock’ or ‘Acid shock’**
 - Snow and ice formed over lakes and rivers have much sulphuric acid. In spring when the snow melts, water enters lakes and rivers, and the sulphuric acid gets mingled into the water bodies making them highly acidic. This condition is commonly referred to as ‘Spring shock’ or ‘Acid shock’. **Hence option (b) is the correct answer.**

- **Gases that cause acid rain**
 - SO_x (Sulphur oxides): Fossil fuel burning, power plants, smelting of metal sulphide ores, industrial sources, industrial production of sulfuric acid in metallurgical, chemical and fertilizer industries volcanoes, seas and oceans, and decomposition of organic matter are the sources of acid rain.
 - NO_x (Nitrogen oxides – NO, NO₂ and N₂O): Fossil fuel burning, lightning, biomass burning, forest fires, oceans, and power plants are sources of nitrogen oxides.
- **Harmful effects of acid rain**
 - Acid precipitation affects both aquatic and terrestrial organisms.
 - It also damages buildings and monuments.
 - Effects on humans
 - Bad smells, reduced visibility; irritation of the skin, eyes and the respiratory tract.
 - Some direct effects include chronic bronchitis, pulmonary emphysema and cancer.
- **Effects on soil**
 - The exchange between hydrogen ions and the nutrient cations like potassium and magnesium in the soil cause leaching of the nutrients, making the soil infertile.
 - An increase in ammonia in the soil due to a decrease in other nutrients decreases the rate of decomposition. The nitrate level of the soil is also found to decrease.
 - The impact of acid rain on soil is less in India; because Indian soils are mostly alkaline, with good buffering ability.
- **Impact on agriculture**
 - Acid precipitation on vegetation shows a reduced rate of photosynthesis.
 - Acid rain retards the growth of crops such as pea, beans, radish, potato, spinach, broccoli and carrots, etc.
 - Plants can absorb cadmium from the acidified soil. High levels of cadmium in plants are injurious for animals and human beings.
- **Effects on aquatic life**
 - Eggs or sperms of fish, frogs and other aquatic organisms are sensitive to pH changes.
 - Acid rain kills their gametes affecting their life cycles and productivity (ecosystem imbalances).
 - Acidic lake waters may kill microbes and turn them unproductive.
 - Acid rain can make metals bound on soils to be released into the aquatic environment.
- **Effect on terrestrial life**
 - Acid rain damages the cuticle of plant leaves and reduces photosynthesis.
 - Acidic medium promotes the leaching of heavy metals like aluminium, lead and mercury. Such metals when percolating into groundwater affect soil microflora/fauna.
 - Other indirect effects of acid rain on wildlife are loss or alteration of food and habitat resources.
- **Effects on microorganisms**
 - pH determines the proliferation of any microbial species.
 - The optimum pH of most bacteria and protozoa is near neutrality.
 - Most fungi prefer an acidic environment.
 - Most blue-green bacteria prefer an alkaline environment.
 - So, microbial species in the soil and water shift from bacteria-bound to fungi-bound.
 - This causes a delay in the decomposition of soil organic material.
- **Effect on buildings, monuments and materials**
 - Many old, historical, ancient buildings and works of art/textile etc. are adversely affected by acid rain.
 - Limestone and marble are destroyed by acid rain. Smoke and soot cover such objects. They slowly dissolve/flake away from the surfaces because of acid fumes in the air.
 - Many buildings/monuments such as the Taj Mahal in Agra have suffered from acid rain (Marble Cancer).

Q 81.B

- Pesticides are substances that are meant to control pests. This includes herbicide, insecticide, nematicide, molluscicide, piscicide, avicide, rodenticide, bactericide, insect repellent, animal repellent, microbicide, fungicide, and lampricide. **A pesticide is a biological or chemical agent that kills animal and plant pests. Hence, statement 1 is not correct.**
 - Algaecides to kill and/or slowing the growth of algae.
 - Antimicrobials to control germs and microbes such as bacteria and viruses.
 - Disinfectants to control germs and microbes such as bacteria and viruses.
 - Fungicides to control fungal problems like molds, mildew, and rust.
 - Herbicides to kill or inhibit the growth of unwanted plants, also known as weeds.
 - Insecticides to control insects.

- Insect Growth Regulators to disrupt the growth and reproduction of insects.
- Rodenticides to kills rodents like mice, rats, and gophers.
- Wood Preservatives to make wood resistant to insects, fungus and other pests
- Biomagnification is any concentration of a toxin, such as **pesticides**, in the tissues of tolerant organisms at successively higher levels in a food chain. **Hence, statement 2 is correct.**
- **BIONEEM** is based on **Azadirachtin** isolated from kernels of neem seeds and this broad-spectrum pesticide provides the most effective, economic and lasting control over major pests of crops. It is the most environment-friendly pesticide, highly biodegradable and **leaves no residue on the foodstuff.** **Hence, statement 3 is correct.**
- Some important pesticides are Endosulfan (insecticide), Sulphur (fungicide), Carbendazim (fungicide) and Dichlorodiphenyltrichloroethane, commonly known as DDT.

Q 82.C

- The process of covering the open surface of the ground by a layer of some external material is called mulching and the material used for covering is called as 'Mulch.' Mulching is usually practiced when cultivating commercially important crops, fruit trees, vegetables, flowers, nursery saplings, etc. It is beneficial in yard gardening, containers & raised beds of home gardens. A mulch is usually, but not exclusively, organic in nature. It may be permanent (e.g. plastic sheeting) or temporary (e.g. bark chips). It may be applied to bare soil or around existing plants.
- Benefits of mulching include:
 - **Shading the soil reduces moisture loss.**
 - **Reduced rain splash reduces pathogen dispersal.**
 - Wind and rain induced soil erosion is reduced.
 - **There is a continuous addition of organic matter to the soil.**
 - **Weeds are suppressed.**
 - Cover protects crops at seedling stage.
 - Soil temperatures are reduced.
- **Hence option (c) is the correct answer.**

Q 83.D

- The Air (Prevention and Control of Pollution) Act was enacted in 1981 to provide for the prevention, control and abatement of air pollution in India. The Act states that the **Central Pollution Control Board** constituted under the **Water (Prevention and Control of Pollution) Act, 1974, shall, without prejudice to the exercise** and performance of its powers and functions under that Act, exercise the powers and perform the functions of the Central Pollution Control Board for the prevention and control of air pollution under this Act. **Hence, statement 1 is not correct.**
 - **Central Pollution Control Board** serves as a field formation and also provides technical services to the Ministry of Environment and Forests of the provisions of the Environment (Protection) Act, 1986. Principal Functions of the CPCB, as spelt out in the Water (Prevention and Control of Pollution) Act, 1974, and the Air (Prevention and Control of Pollution) Act, 1981, (i) to promote cleanliness of streams and wells in different areas of the States by prevention, control and abatement of water pollution, and (ii) to improve the quality of air and to prevent, control or abate air pollution in the country.
 - **Air Quality Monitoring** is an important part of the air quality management. The National Air Monitoring Programme (NAMP) has been established with objectives to determine the present air quality status and trends and to control and regulate pollution from industries and other source to meet the air quality standards. It also provides background air quality data needed for industrial siting and towns planning.
- One of the **disadvantages** of the act is that any **pollutant into the atmosphere by ship or aircraft** doesn't come under the purview of the act. **Hence, statement 2 is not correct.**

Q 84.C

- **Why in the news?**
 - The Prime Minister launched the year-long 125th birth anniversary celebration of legendary freedom fighter Alluri Sitarama Raju in Bhimavaram, Andhra Pradesh.
 - The Prime Minister informed that the 125th birth anniversary of Alluri Sitarama Raju Garu and the 100th anniversary of Rampa Kranti will be celebrated throughout the year.
 - Restoration of his birth place at Pandrangi, renovation of Chintapalli police station, construction of Alluri Dhyana Mandir at Mogallu, these works are symbols of the spirit of Amrit Mahotsav.

- **Alluri Sitarama Raju (4 July 1897 or 1898 – 7 May 1924) was an Indian revolutionary who waged an armed campaign against British colonial rule in India.**
- **Born in present-day Andhra Pradesh,** he became involved in opposing the British in response to the 1882 Madras Forest Act, which effectively restricted the free movement of Adivasis (tribal communities) in their forest habitats and prevented them from practicing a traditional form of agriculture known as podu.
- Rising discontent towards the British led to the Rampa Rebellion of 1922, in which he played a major part as a leader. Mustering a force combined of Adivasis, farmers and sympathizers to their cause, he engaged in guerilla campaigns against the British colonial authorities in the border regions of the Madras Presidency part of the districts of East Godavari and Visakhapatnam (now part of Alluri Sitharama Raju district).
- **He was nicknamed "Manyam Veerudu" (transl. Hero of the Jungle) by local villagers for his heroic exploits.**
- **Hence option (c) is the correct answer.**

Q 85.D

- Ecological Pyramid is a graphical depiction that is meant to illustrate the relationship between different living organisms at different trophic levels in an ecosystem. In an ecological pyramid there are three types of ecological pyramids.
- **Pyramid of Numbers** represent the total numbers of individuals(population) present in each trophic level.
- **Pyramid of biomass** indicates the total mass of organisms in a particular trophic level.
- **Pyramid of energy** is an upright pyramid that illustrates the flow of energy from producers to consumers. Furthermore, it indicates the actual role played by various organisms in energy transfer. Energy pyramids indicate how much energy is required in the next trophic level as it flows upwards.
- Any calculations of energy content, biomass, or numbers has to include all organisms at that trophic level. No generalizations we make will be true if we take only a few individuals at any trophic level into account. Also a given organism may occupy more than one trophic level simultaneously. One must remember that the trophic level represents a functional level, not a species as such. A given species may occupy more than one trophic level in the same ecosystem.
- In most ecosystems, all the pyramids, of number, of energy and biomass are upright, i.e., producers are more in number and biomass than the herbivores, and herbivores are more in number and biomass than the carnivores. Also energy at a lower trophic level is always more than at a higher level.
- **There are exceptions to this generalization. The pyramid of biomass in the sea is also generally inverted because the biomass of fishes far exceeds that of phytoplankton.**
- **The pyramid of energy is always upright, and can never be inverted, because when energy flows from a particular trophic level to the next trophic level, some energy is always lost as heat at each step. Each bar in the energy pyramid indicates the amount of energy present at each trophic level in a given time or annually per unit area.**
- There are certain limitations of ecological pyramids such as it does not take into account the same species belonging to two or more trophic levels. It assumes a simple food chain, something that almost never exists in nature; it does not accommodate a food web. **Moreover, saprophytes are not given any place in ecological pyramids even though they play a vital role in the ecosystem.**
- **Hence, all three statements are correct.**

Q 86.D

- **Invasive Species:** Plants, animals, fungi, or microorganisms that spread rapidly and cause harm to other species are called Invasive Species. Invasive species are sometimes so harmful and damaging that they threaten an entire ecosystem. An “invasive species” is defined as a species that is -
 - Non-native (or alien) to an ecosystem, and
 - Whose introduction causes or is likely to cause economic or environmental harm or harm to human health.
- In fact invasive species may be the species of some foreign organisms that cause harm to organisms of a new ecosystem when introduced in it. When alien species are introduced unintentionally or deliberately for whatever purpose, some of them turn invasive, and cause decline or extinction of indigenous species.
- The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fish in the lake. You must be familiar with the environmental damage caused and the threat posed to our native species by invasive weed species like **carrot grass (Parthenium), Lantana, and water hyacinth (Eichhornia).** **Hence, option (d) is the correct answer.**

- The recent illegal introduction of the **African catfish *Clarias gariepinus*** for aquaculture purposes is posing a threat to the indigenous catfishes in our rivers. ***Euglandina rosea* (Rosy wolf snail) and Tasmanian Brush-tail Possum (*Trichosurus vulpecula*)** are two other examples of invasive species.

Q 87.D

- **Extinction** occurs when species are diminished because of environmental forces (habitat fragmentation, global change, natural disaster, overexploitation of species for human use) or because of evolutionary changes in their members (genetic inbreeding, poor reproduction, decline in population numbers).
- Ecologists estimate that the present-day extinction rate is 1,000 to 10,000 times the background extinction rate (between one and five species per year) because of deforestation, habitat loss, overhunting, pollution, climate change, and other human activities.
- Some examples of recent extinctions include the **dodo (Mauritius), quagga (Africa), thylacine (Australia), Steller's Sea Cow (Russia) and three subspecies (Bali, Javan, Caspian) of tiger**. Hence, **option (d) is the correct answer.**
- **Some of the animals that have gone extinct in recent times are**
 - Pink-headed duck
 - Himalayan quail
 - Japanese sea lion
 - Guam flying fox
 - Caspian Tiger
 - Yangtze River Dolphin
 - Passenger Pigeon
 - The Quagga
 - The Golden Toad
 - West African Black Rhino
 - Javan Tiger

Q 88.C

- In the Soviet period, the Caspian Sea was practically an inland water basin within the Soviet Union's borders and washed off the coast of Iran only to the south. Until 1992, the status of the Caspian was regulated by the Soviet-Iranian treaties. **After the collapse of the USSR, emerging independent states, such as Azerbaijan, Kazakhstan, and Turkmenistan, raised the issue of dividing the Caspian Sea.**
- Although meetings with the representatives of the coastal states on how to settle the status of the Caspian Sea aimed to make a final decision, this was not possible, and there was a need to discuss the issue at the Summit. During these talks, the meeting held in Ashgabat on April 22–23, 2002 was of historical importance.
- On April 23–24, 2002, the **first summit of the Caspian states** was held in Ashgabat, Turkmenistan. The Summit of the heads of Caspian littoral states with the participation of Presidents Heydar Aliyev (Azerbaijan), Saparmurat Niyazov (Turkmenistan), Mohammad Khatami (Iran), Nursultan Nazarbayev (Kazakhstan) and Vladimir Putin (Russia) was the first summit.
- The **second summit of the Caspian littoral states** was held in Tehran on October 16, 2007. At the end of the Summit, a Declaration was signed by the heads of the Caspian littoral states. The Declaration consisted of 25 items.
- The **3rd Caspian Summit** was held on November 18, 2010, in Baku. The heads of Caspian littoral states signed an agreement on the cooperation on the security in the Caspian Sea.
- On September 29, 2014, the **fourth Caspian Summit** was held in Astrakhan. The Caspian littoral states discussed the legal status, security, biological resources, and environmental problems of the Caspian Sea.
- The **5th Caspian Summit** was held in Aktau in 2018. At this summit, the parties signed a Convention on the Legal Status of the Caspian Sea.
- **The 6th Caspian Summit was held in Ashgabat on June 29, 2022. The presidents of Russia, Azerbaijan, Iran, Kazakhstan, and Turkmenistan discussed topical issues of cooperation in the Caspian Sea in relation to various spheres, as well as the implementation of resolutions made during the previous meetings of the heads of the Caspian “five”.**
- Hence, **option (c) is the correct answer.**

Q 89.D

- **Recent context: The Indian Space Research Organisation (ISRO) achieved the feat of successfully launching the PSLV Orbital Experimental Module or ‘POEM’.**
- **POEM (PSLV Orbital Experimental Module) is an experimental mission by ISRO which performs in-orbit scientific experiments during the fourth stage of the Polar Satellite Launch Vehicle (PSLV) launch vehicle as an orbital platform. Hence statement 1 is not correct.**

- The PSLV is a four-stage rocket where the first three spent stages fall back into the ocean, and the final stage (PS4) — after launching the satellite into orbit — ends up as space junk.
- POEM has a dedicated Navigation Guidance and Control (NGC) system for attitude stabilization, which stands for controlling the orientation of any aerospace vehicle within permitted limits. The NGC will act as the platform's brain to stabilize it with specified accuracy. **It is the first time that the (fourth stage) PS4 stage would orbit the earth as a stabilized platform.**
- **POEM will derive its power from solar panels** mounted around the PS4(fourth stage) tank, **and a Li-Ion battery.** It will navigate using Four Sun Sensors, a Magnetometer, Gyros & NavIC. **Hence statement 2 is not correct.**

Q 90.D

- **Biopesticides:** They are biological agents that are helpful in controlling plant pests. There are basically two types of bio-pesticides: Bio-herbicides and bio-insecticides. Bio-herbicides are living organisms that destroy herbs. These may be insects that feed on specific weeds or microorganisms that may cause diseases in weeds. Here are some examples –
 - **Mycoherbicide:** This is derived from a fungus called *Phytophthora palmivora*. It controls the growth of milkweed vines in citrus.
 - **Cochineal insect (Cactoblastic cactorum):** This insect eats away cactus plants. This is being introduced in controlling the overgrowth of cacti in India and Australia.
 - **Pyrethrum (pyrethins), rotenone or ryania (botanical insecticides), or minerals, such as boric acid, cryolite, or diatomaceous earth.**
 - **Bacillus Thuringiensis** is a naturally occurring soil bacteria that is toxic to the larvae of several species of insects but not toxic to non-target organisms. **Hence option (d) is the correct answer.**
- **Glyphosate** is a broad-spectrum systemic herbicide and crop desiccant. It is an organophosphorus compound, specifically a phosphonate, which acts by inhibiting the plant enzyme 5-enolpyruvylshikimate-3-phosphate synthase.
- Recently, transgenic plants of vegetables like tomato and tobacco have been developed through genetic engineering. These plants are so engineered genetically that they can develop resistance against herbicides. Garlic, Cloves, Cedarwood oil, etc. have been tested indigenously and have been found useful in repelling certain insect- pests. Besides these, Pheromones like Xanthenes and Plant Hormones like Auxins, gibberellins, etc. are also used as bio-pesticides.
- **Bio-insecticides:** The natural enemies of particular insects and certain other substances that are used for killing or repelling insect pests in a particular area are called bio-insecticides. There are the following categories of bio-insecticides –
 - Disease-causing organisms (pathogens), parasites, and predators.
 - Pheromones or insect hormones.
 - Natural Insecticides like tobacco extract, neem oil, and specific bacterial toxins (called Thuriosides).
 - Viral bio-insecticides like Bioculoviruses affect insect pests.
 - Fungal agents or different fungi are useful in controlling insect pests.
 - Insect insecticides like Lady Bird and Praying Mantis eat away many types of insects.

Q 91.D

- CITES (the Convention on International Trade in Endangered Species of Wild Fauna and Flora) is an international agreement between governments. Its aim is to ensure that international trade in specimens of wild animals and plants does not threaten the survival of the species. CITES was drafted as a result of a resolution adopted in 1963 at a meeting of members of IUCN (**The World Conservation Union**). **Hence, statement 1 is correct.**
- CITES is an international agreement to which States and regional economic integration organizations adhere voluntarily. States that have agreed to be bound by the Convention ('joined' CITES) are known as Parties. **Although CITES is legally binding on the Parties – in other words they have to implement the Convention – it does not take the place of national laws.** Rather it provides a framework to be respected by each Party, which has to adopt its own domestic legislation to ensure that CITES is implemented at the national level. **CITES Secretariat is administered by UNEP** (The United Nations Environment Programme) and is located at Geneva, Switzerland. **Hence, statement 2 is correct.**
- The species covered by **CITES are listed in three Appendices**, according to the degree of protection they need. There are special rules in these cases and a permit or certificate will generally still be required. Anyone planning to import or export/re-export specimens of a CITES species should contact the national CITES Management Authorities of the countries of **import and export/re-export for information on the rules that apply.** **Hence, statement 3 is correct.**

- **Appendices I, II and III to the Convention** are lists of species afforded different levels or types of protection from over-exploitation. **Appendix I** lists species that are the most endangered among CITES-listed animals and plants. **Appendix II** lists species that are not necessarily now threatened with extinction but that may become so unless trade is closely controlled. **Appendix III** is a list of species included at the request of a Party that already regulates trade in the species and that needs the cooperation of other countries to prevent unsustainable or illegal exploitation.

Q 92.B

- **Recent context: The Ministry of Women and Child Development has issued detailed guidelines for the 'Mission Shakti' scheme.** These guidelines shall be implemented from 1st April 2022. 'Mission Shakti' was launched during the 15th Finance Commission period 2021-22 to 2025-26. **Mission Shakti is an integrated women empowerment programme is launched as an umbrella scheme for the safety, security and empowerment of women for implementation.**
- The components of Mission Shakti have been designed in such a way that it takes care of the women's needs on a life cycle continuum basis. **Mission Shakti has two sub-schemes - 'sambal' and 'Samarthya'. While the "Sambal" sub-scheme is for the safety and security of women, the "Samarthya" sub-scheme is for the empowerment of women.**
- The components of the 'sambal' subscheme consist of erstwhile schemes of One Stop Centre (OSC), Women Helpline (WHL), Beti Bachao Beti Padhao (BBBP) with a new component of Nari Adalats - women's collectives to promote and facilitate alternative dispute resolution and gender justice in society and within families. **Hence statement 2 is correct.**
- **'Samarthya' for Empowerment of Women.** It consists of erstwhile schemes of Ujjwala, Swadhar Greh and Working Women Hostel that have been included with modifications. In addition, the existing schemes of the National Creche Scheme for children of working mothers and Pradhan Mantri Matru Vandana Yojana (PMMVY) under the umbrella of Integrated Child Development Services ICDS have now been included in Samarthya. A new component of Gap Funding for Economic Empowerment has also been added to the Samarthya Scheme.
- The sub-scheme of 'Sambal' will be implemented as a Centrally Sponsored scheme with 100% central funding from the Nirbhaya Fund/ MWCD budget with provision for direct release of funds to the District Collector or the concerned Directorate/Commissionerate in State/ UT. The sub-scheme 'Samarthya' will be implemented as a Centrally Sponsored Scheme with a funding ratio of 60:40 between Centre and State Governments / UTs with the legislature, except North East & Special Category States / UTs with legislature where the fund ratio will be 90:10. For UTs without legislature, 100% funding will be provided by the Central Government. **Hence statement 1 is not correct.**

Q 93.B

- **Indian Institute of Technology Madras (IIT Madras) Researchers have developed an Artificial Intelligence-based tool, 'PIVOT', that can predict cancer-causing genes in an individual.** This tool will ultimately help in devising personalized cancer treatment strategies.
- The prediction is based on a model that utilizes information on mutations, expression of genes, and copy number variation in genes and perturbations in the biological network due to an altered gene expression.
- **The tool is based on a machine learning model that classifies genes as tumor suppressor genes, oncogenes, or neutral genes.** The tool was able to successfully predict both the existing oncogenes and tumor-suppressor genes like TP53, and PIK3CA, among others, and new cancer-related genes such as PRKCA, SOX9, and PSMD4.
- Although there are tools available to identify personalized cancer genes, they use unsupervised learning and predict based on the presence and absence of mutations in cancer-related genes. This study, however, is the first one to use supervised learning and takes into account the functional impact of mutations while making predictions.
- **Hence option (b) is the correct answer.**

Q 94.D

- Living organisms try to maintain the constancy of their internal environment (a process called homeostasis) despite varying external environmental conditions that tend to upset their homeostasis.
- Abiotic conditions of many habitats vary drastically over time, to deal with this and maintain homeostasis, organisms living in such habitats cope with the stressful conditions by various mechanisms.
- Many organisms if unable to migrate from harsh conditions, avoid the stress by escaping in time. **The familiar case of bears going into hibernation during winter is an example of escape in time. Some snails and fish go into aestivation to avoid summer-related problems-heat and desiccation.**

- Under unfavorable conditions, many zooplankton species in lakes and ponds are known to enter diapause, a stage of suspended development.
- Hence, option (d) is the correct answer.

Q 95.D

- Scientists from the Wadia Institute of Himalayan Geology (WIHG), an autonomous institute of the Department of Science and Technology (DST) found fossils of the new genus and species of treeshrew (known as *Sivatupaia ramnagarensis* n. gen. n. sp) i.e. fossils of a small mammal resembling squirrels from the middle Miocene (extends from about 23.03 to 5.333 million years ago) site of Ramnagar in Jammu and Kashmir.
 - It can provide a precise age estimate for the locality.
 - This treeshrew currently represents the oldest record of fossil tupaiids in the Siwaliks, extending their time range by 2.5–4.0 Million Years in the region and can help provide a more precise age estimate for this Ramnagar locality lying in the Udhampur District (UT-Jammu & Kashmir).
 - Siwalik sediments document the evolution of many mammalian groups from the middle Miocene through the Pleistocene including treeshrews, hedgehogs, and other small mammals. Treeshrews, in particular, are very rare elements of the fossil record, with only a few species known throughout the entire Cenozoic era.
- Hence option (d) is the correct answer.

Q 96.D

- The concept of sustainable development was described by the 1987 Bruntland Commission Report as “development that meets the needs of the present without compromising the ability of future generations to meet their own needs.”
- There are four dimensions to sustainable development – society, environment, culture and economy – which are intertwined, not separate. Sustainability is a paradigm for thinking about the future in which environmental, societal and economic considerations are balanced in the pursuit of improved quality of life. For example, a prosperous society relies on a healthy environment to provide food and resources, safe drinking water and clean air for its citizens.
- The strategies for sustainable development are as follows
 - Use of Non-conventional Sources of Energy
 - Use of Compressed Natural Gas (CNG)
 - Biocomposting
 - Solar Power through Photovoltaic Cells
 - LPG, Gobar Gas in Rural Areas
 - Wind Power
 - Mini-hydel Plants
- Hence, option (d) is the correct answer.

Q 97.D

- While considering the various alternatives available to organisms for coping with extremes in their environment, some are able to respond through certain physiological adjustments while others do so behaviourally (migrating temporarily to a less stressful habitat). These responses are also actually, their adaptations. Adaptation is any attribute of the organism (morphological, physiological, behavioural) that enables the organism to survive and reproduce in its habitat. Many adaptations have evolved over a long evolutionary time and are genetically fixed.
- In the absence of an external source of water, the kangaroo rat in North American deserts is capable of meeting all its water requirements through its internal fat oxidation (in which water is a by-product). It also has the ability to concentrate its urine so that minimal volume of water is used to remove excretory products.
- Many desert plants have a thick cuticles on their leaf surfaces and have their stomata arranged in deep pits to minimize water loss through transpiration. They also have a special photosynthetic pathway (CAM) that enables their stomata to remain closed during daytime. Some desert plants like *Opuntia*, have no leaves – they are reduced to spines–and the photosynthetic function is taken over by the flattened stems.
- Hence, option (d) is the correct answer.

Q 98.B

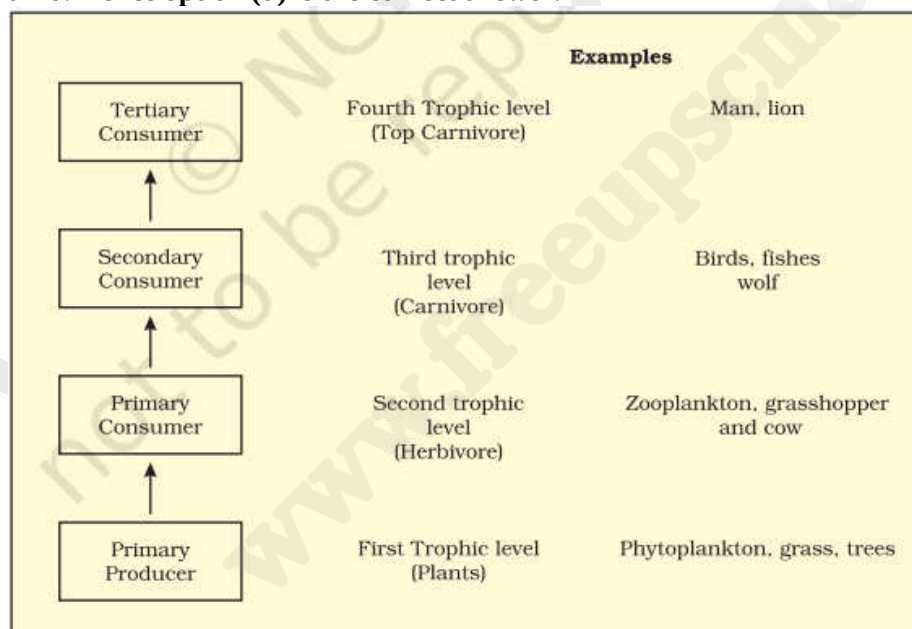
- **The Montreal Protocol on Substances that Deplete the Ozone Layer** is the landmark multilateral environmental agreement that regulates the production and consumption of nearly 100 man-made chemicals referred to as ozone depleting substances (ODS).
- **Recognising the deleterious affects of ozone depletion, an international treaty, known as the Montreal Protocol, was signed at Montreal (Canada) in 1987** (effective in 1989) to control the emission of ozone depleting substances. Subsequently many more efforts have been made and protocols have laid down definite roadmaps, separately for developed and developing countries, for reducing the emission of CFCs and other ozone depleting chemicals. **Hence, option (b) is the correct answer.**
- Multilateral Fund for the Implementation of the Montreal Protocol was established in 1991. The Multilateral Fund's activities are implemented by four international agencies - UN Environment Programme (UNEP), UN Development Programme (UNDP), UN Industrial Development Organisation (UNIDO) and the World Bank - as well as bilateral agencies of non-Article 5 countries.
- Parties to the Montreal Protocol reached agreement at their 28th Meeting of the Parties in Kigali, Rwanda to phase-down HFCs. Countries agreed to add HFCs to the list of controlled substances, and approved a timeline for their gradual reduction by 80-85 per cent by the late 2040s. Developing countries will follow with a freeze of HFCs consumption levels in 2024 and in 2028 for some nations.

Q 99.B

- **Statement 1 is not correct:** Each trophic level has a certain mass of living material at a particular time called as the standing crop. It is also defined as the total amount or weight, or energy content, of (a portion of) organisms existing in a specific area at a particular time.
- **Statement 2 is correct: standing crop is measured as the mass of living organisms (biomass) or the number in a unit area.**
- The biomass of a species is expressed in terms of fresh or dry weight. Measurement of biomass in terms of dry weight is more accurate.

Q 100.D

- Organisms occupy a place in the natural surroundings or in a community according to their feeding relationship with other organisms. Based on the source of their nutrition or food, organisms occupy a specific place in the food chain that is known as their trophic level. Producers belong to the first trophic level, herbivores (primary consumer) to the second and carnivores (secondary consumer) to the third. **Hence option (d) is the correct answer.**



Diagrammatic representation of trophic levels in an ecosystem